PMBOK® Guide – Eighth Edition & PMP® — Comprehensive Question Bank (with Answers)

Purpose. This question bank checks your knowledge of the *whole* body of material in the Study Guide: the Standard (value delivery, environment, principles), the seven performance domains and their 40 processes, quality, agile and hybrid delivery, people skills, the business environment, tailoring, the toolbox of models/methods/artifacts, and calculations. The final parts mix topics in realistic scenarios, the way the exam does.

How to use it

1. Work through one part at a time without notes. Write down your answers.
2. Check the answers at the end of the part. Each answer has a short explanation — read it even when you were right.
3. Record your score per part in the table below. Anything under **80%** shows a topic to revisit in the Study Guide (the chapter references are given in the table).
4. Come back to the parts you scored lowest on a week later.

Part	Topic	Study Guide chapters	Your score
1	Foundations, value delivery and principles	2, 3, 4	
2	Life cycles, focus areas and processes	5, 6	
3	Governance	7	
4	Scope and quality	8, 9	
5	Schedule	10	
6	Finance	11	
7	Stakeholders and communications	12	
8	Resources and procurement	13	
9	Risk	14	
10	Agile and hybrid delivery	15	
11	Leading people and ethics	16	
12	Business environment, tailoring and toolbox	17, 18, 19	
13	Calculations workout	20	
14	Integrated scenarios	All	

Question types. Most questions have one best answer. Multiple-response questions tell you how many options to choose; matching questions ask you to pair items; fill-in-the-blank questions need a number. As on the real exam, a multiple-response or matching question counts as correct only if every part is right.

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- Part 4 — Scope and Quality (Q71–Q95) · answers
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- Part 7 — Stakeholders and Communications (Q136–Q155) · answers
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- Part 10 — Agile and Hybrid Delivery (Q196–Q219) · answers
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- Part 12 — Business Environment, Tailoring and Toolbox (Q245–Q269) · answers
- Part 13 — Calculations Workout (Q270–Q284) · answers
- Part 14 — Integrated Scenarios (Q285–Q304) · answers

Part 1 — Foundations, Value Delivery and Principles

1. Which characteristic distinguishes a project from operations?

- **A.** Projects maintain existing capabilities, while operations create new ones
- **B.** Operations have formal stakeholders, while projects only have team members
- **C.** Projects are temporary and produce unique results; operations are ongoing and repetitive
- **D.** Projects always have larger budgets and more senior sponsors than any ongoing operations

2. A city launches several related projects — new bus lanes, a ticketing app and a park-and-ride facility — coordinated to reduce traffic congestion. How is this collection best described?

- **A.** Operations
- **B.** A program
- **C.** A single project with three phases
- **D.** A portfolio

3. Which statement best describes the relationship between outputs, outcomes and benefits?

- **A.** Outputs are what is produced; outcomes result from using them; benefits are the gains that follow
- **B.** Benefits are produced first, then converted into outputs that create the outcomes stakeholders see
- **C.** The three terms are interchangeable labels for the deliverables accepted at project closure
- **D.** Outcomes are the deliverables, outputs are the gains, and benefits are the activities producing them

4. In the system for value delivery, what is the primary purpose of portfolio management?

- **A.** Running day-to-day operations so that project deliverables are sustained
- **B.** Approving each project's change requests to keep baselines under control
- **C.** Coordinating the detailed schedules of related projects to share resources efficiently
- **D.** Selecting and prioritising investments to achieve strategic objectives

5. Which of the following is an internal enterprise environmental factor?

- **A.** Interest rates set by the central bank
- **B.** Weather conditions at the site
- **C.** The organisation's geographic distribution of facilities and staff
- **D.** Industry standards published by a professional association for its members

6. A project manager uses templates and historical estimates from a company database to plan a new project. These are best classified as:

- **A.** Work performance data
- **B.** Organisational process assets
- **C.** Enterprise environmental factors
- **D.** Project deliverables

7. In a strong matrix organisation, which statement is typically true?

- **A.** The project manager acts as an expediter without authority over resources or budget
- **B.** Team members report only to the project manager and have no functional manager
- **C.** The functional manager controls the budget and the project manager coordinates part-time
- **D.** The project manager has moderate to high authority and is assigned full-time

8. Who usually issues the project charter?

- **A.** The procurement department, on behalf of the PMO
- **B.** The project team
- **C.** The sponsor or project initiator
- **D.** The project manager

9. Which two items are commonly found in a business case? (Choose two.)

- **A.** Analysis of the business need and recommended option
- **B.** Daily stand-up notes
- **C.** The team's ground rules
- **D.** Cost-benefit analysis with financial measures such as NPV
- **E.** Detailed activity durations

10. A team member asks why the project bothers to measure benefits after go-live, since the project will have closed. What is the best explanation?

- **A.** Measuring benefits after go-live is required only when the project used an agile approach
- **B.** Many benefits appear only after delivery, so measuring them confirms the value created
- **C.** It is not needed, because success is fully determined by scope, time and cost at closure
- **D.** Benefit tracking after closure is mainly needed to calculate the team's performance bonuses

11. Which principle is most directly applied when a project manager considers how a design decision will affect maintenance costs and energy use over the product's 20-year life?

- **A.** Integrate sustainability within all project areas
- **B.** Be an accountable leader
- **C.** Embed quality, because energy use is a defect in the delivered product
- **D.** Build an empowered culture

12. A project manager notices that optimising the project's schedule would overload a shared testing team needed by two other critical projects. Which principle should guide the response?

- **A.** Build an empowered culture
- **B.** Focus on value only for this project
- **C.** Adopt a holistic view
- **D.** Embed quality into deliverables

13. Which behaviour best demonstrates the principle "Build an empowered culture"?

- **A.** Limiting stakeholder involvement to formal monthly reports to keep the team focused
- **B.** Inviting people to raise concerns and letting the team decide within agreed boundaries
- **C.** Requiring the project manager's approval for decisions so that accountability stays clear
- **D.** Publicly naming people who make mistakes so that the whole team learns from them

14. A senior developer proposes adding extra features that no one has requested, arguing they will "add quality". Which principle helps the project manager respond?

- **A.** Focus on value — any additional feature increases the value delivered to the customer
- **B.** Adopt a holistic view — a larger scope improves the system as a whole
- **C.** Be an accountable leader — the senior developer should decide on all technical scope alone
- **D.** Embed quality — quality means meeting requirements and fitness for use, not extra features

15. How many principles does the Eighth Edition define, and how many did the Seventh Edition define?

- **A.** Seven in the Eighth Edition; ten in the Seventh Edition
- **B.** Twelve in the Eighth Edition; six in the Seventh Edition
- **C.** Eight in both editions
- **D.** Six in the Eighth Edition; twelve in the Seventh Edition

16. Which side of the PMI Talent Triangle covers understanding the organisation's strategy, finances and market?

- **A.** Ways of Working
- **B.** Technical Leadership
- **C.** Power Skills
- **D.** Business Acumen

17. What does it mean that the PMBOK Guide is "non-prescriptive"?

- **A.** It describes good practice that teams adapt to their context instead of mandating procedures
- **B.** It applies only to predictive projects, while agile projects follow the Agile Practice Guide
- **C.** It prescribes standard templates that every organisation must adopt without changes
- **D.** It is a theoretical reference that is not intended to be used on real projects

18. A government agency must deliver a new tax portal because of a new law. Which category of project driver does this represent?

- **A.** Regulatory, legal or social requirement
- **B.** Stakeholder request or need
- **C.** Business or technological strategy
- **D.** Creating, improving or fixing a product

19. Which role is usually responsible for prioritising investments across many projects when resources are scarce?

- **A.** The PMO's quality assurance analysts
- **B.** Each project manager, negotiating independently with the sponsors
- **C.** The scrum master of the largest team
- **D.** The portfolio manager or portfolio governance board

20. Match each term with its description.

(Matching: pair every numbered item with one lettered option.)

Item	Option
1. Program	**A.** Ongoing work that sustains the organisation

Item	Option
2. Portfolio	**B.** Components managed as a group to achieve strategic objectives
3. Operations	**C.** An artifact that may be created or changed by several projects over its life cycle
4. Product	**D.** Related components coordinated to obtain benefits

21. A project manager discovers that a vendor's materials contain a substance that violates environmental regulations, although it is cheaper. What should the project manager do?

- **A.** Continue with the materials but record the issue in the risk register for later review
- **B.** Ask the vendor to provide documentation that classifies the substance as acceptable
- **C.** Use the materials, since the savings were approved in the budget and the client wants low cost
- **D.** Stop using the materials, inform the right stakeholders and source a compliant alternative

22. Which document describes target benefits, the timeframe for realising them and who is accountable for them?

- **A.** Communications management plan
- **B.** Issue log
- **C.** Benefits management plan
- **D.** Project scope statement

23. What is the main reason projects should report honestly upward through the value delivery system?

- **A.** Operations teams use project reports to approve the project budget each month
- **B.** Honest reports are required mainly to pass audits at the end of the project
- **C.** Leadership needs accurate information to adjust strategy and investments
- **D.** Status reports determine the project manager's annual performance rating

24. A project manager is told to "act as an integrator". What does this mean?

- **A.** Bringing plans, work, people and changes together into a coherent whole
- **B.** Approving every change request quickly so that the plan stays integrated
- **C.** Personally performing the technical integration of all system components
- **D.** Delegating the alignment of plans to the sponsor and functional managers

25. Which statement about value is correct according to the Standard?

- **A.** Value can be tangible or intangible and may be realised during or after the project
- **B.** Value is measured only by profit and is confirmed when the final invoice is paid
- **C.** Value equals the approved budget, because it reflects what stakeholders will pay
- **D.** Value is realised at project closure, when all deliverables are formally accepted by the customer

Answers — Part 1 (Foundations, Value Delivery and Principles)

Q1 — Answer: C

Correct: C. Projects are temporary and produce unique results; operations are ongoing and repetitive

Projects are temporary endeavours that create unique products, services or results; operations sustain the business through repetitive work. Projects usually change operations.

Q2 — Answer: B

Correct: B. A program

Related projects coordinated to achieve benefits (reduced congestion) that would not be achieved by managing them separately form a program.

Q3 — Answer: A

Correct: A. Outputs are what is produced; outcomes result from using them; benefits are the gains that follow

Activities create outputs; using the outputs creates outcomes; outcomes produce benefits; together they contribute to value.

Q4 — Answer: D

Correct: D. Selecting and prioritising investments to achieve strategic objectives

Portfolios align investments with strategy, balance risk and allocate resources. Detailed scheduling and change approval occur at project level.

Q5 — Answer: C

Correct: C. The organisation's geographic distribution of facilities and staff

Geographic distribution of facilities is internal to the organisation and not under the team's immediate control. Industry standards, interest rates and weather are external.

Q6 — Answer: B

Correct: B. Organisational process assets

Templates, historical information and knowledge repositories belong to the performing organisation and are OPAs.

Q7 — Answer: D

Correct: D. The project manager has moderate to high authority and is assigned full-time

In a strong matrix, PM authority and resource availability are moderate to high, and the PM is full-time. Team members still have functional managers.

Q8 — Answer: C

Correct: C. The sponsor or project initiator

The charter is issued by someone external to the project at a level able to fund and authorise it — usually the sponsor.

Q9 — Answer: A, D

Correct: A. Analysis of the business need and recommended option / D. Cost-benefit analysis with financial measures such as NPV

A business case justifies the investment: it describes the need, analyses options, recommends one and includes cost-benefit analysis. Activity durations and team artifacts belong elsewhere.

Q10 — Answer: B

Correct: B. Many benefits appear only after delivery, so measuring them confirms the value created

Value is often realised after the project closes. The benefit owner tracks benefits to confirm the business case and inform future decisions.

Q11 — Answer: A

Correct: A. Integrate sustainability within all project areas

Considering long-term environmental and economic impacts over the product life cycle reflects sustainability, while also reflecting a holistic view.

Q12 — Answer: C

Correct: C. Adopt a holistic view

A holistic view considers the project as part of a larger system, including shared resources and other initiatives.

Q13 — Answer: B

Correct: B. Inviting people to raise concerns and letting the team decide within agreed boundaries

Empowered cultures rely on psychological safety, trust and delegated decision-making with clear boundaries.

Q14 — Answer: D

Correct: D. Embed quality — quality means meeting requirements and fitness for use, not extra features

Gold plating is not quality. Quality is defined by requirements and real stakeholder needs; extra features add cost and risk without agreed value.

Q15 — Answer: D

Correct: D. Six in the Eighth Edition; twelve in the Seventh Edition

The Eighth Edition consolidated the Seventh Edition's twelve principles into six.

Q16 — Answer: D

Correct: D. Business Acumen

Business Acumen connects projects to strategy and value through knowledge of the business, finance, markets and regulations.

Q17 — Answer: A

Correct: A. It describes good practice that teams adapt to their context instead of mandating procedures

The Guide describes what is generally good practice; the team tailors processes, methods and artifacts to the context.

Q18 — Answer: A

Correct: A. Regulatory, legal or social requirement

Projects initiated to comply with new laws or regulations fall under regulatory, legal or social requirements.

Q19 — Answer: D

Correct: D. The portfolio manager or portfolio governance board

Resource conflicts across projects that affect strategy are resolved at portfolio level, which prioritises investments.

Q20 — Answer: 1→D, 2→B, 3→A, 4→C

Correct pairs: 1. Program → Related components coordinated to obtain benefits; 2. Portfolio → Components managed as a group to achieve strategic objectives; 3. Operations → Ongoing work that sustains the organisation; 4. Product → An artifact that may be created or changed by several projects over its life cycle

Programs coordinate for benefits; portfolios align investments to strategy; operations sustain the business; products persist over a life cycle that can span many projects.

Q21 — Answer: D

Correct: D. Stop using the materials, inform the right stakeholders and source a compliant alternative

Accountable leadership, sustainability and compliance require transparency and adherence to laws. Cost savings never justify illegal or unethical actions.

Q22 — Answer: C

Correct: C. Benefits management plan

The benefits management plan defines benefits, metrics, timeframe, benefit owner, assumptions and risks.

Q23 — Answer: C

Correct: C. Leadership needs accurate information to adjust strategy and investments

The value delivery system depends on feedback loops; inaccurate status prevents good portfolio and strategic decisions.

Q24 — Answer: A

Correct: A. Bringing plans, work, people and changes together into a coherent whole

Integration is the PM's core role: aligning processes, knowledge, context and complexity to deliver outcomes.

Q25 — Answer: A

Correct: A. Value can be tangible or intangible and may be realised during or after the project

Value includes financial and non-financial benefits such as reputation or public good, and is often realised progressively or after closure.

Part 2 — Life Cycles, Focus Areas and Processes

26. A team prototypes a user interface three times based on user feedback before building it, and then delivers the whole product at once. Which statement best describes this life cycle?

- **A.** Continuous delivery
- **B.** Predictive with no feedback
- **C.** Iterative development with a single delivery
- **D.** Incremental delivery with multiple releases to users

27. Which focus area contains the largest number of processes in the Eighth Edition?

- **A.** Initiating
- **B.** Monitoring and Controlling
- **C.** Planning
- **D.** Executing

28. Which of the following processes belongs to the Executing focus area?

- **A.** Plan Risk Responses
- **B.** Perform Risk Analysis
- **C.** Monitor Risks
- **D.** Implement Risk Responses

29. Which process belongs to the Monitoring and Controlling focus area?

- **A.** Elicit and Analyze Requirements
- **B.** Develop Scope Structure
- **C.** Validate Scope
- **D.** Define Scope

30. In which performance domain is Plan Sourcing Strategy located?

- **A.** Governance
- **B.** Risk
- **C.** Resources
- **D.** Finance

31. Which Eighth Edition process replaced "Direct and Manage Project Work"?

- **A.** Monitor and Control Project Performance
- **B.** Manage Quality Assurance
- **C.** Manage Project Execution
- **D.** Lead the Team

32. A phase gate review concludes that the project should continue only if the scope is reduced. What decision outcome is this?

- **A.** Continue with modifications
- **B.** End the project
- **C.** Continue without change

- **D.** Repeat the phase

33. Which two statements about project documents and the project management plan are correct? (Choose two.)

- **A.** The stakeholder register is part of the cost baseline
- **B.** The issue log can be updated without a formal change request
- **C.** The risk register is a component of the scope baseline
- **D.** Project documents can never be changed after planning
- **E.** Changes to the schedule baseline require formal change control

34. Which output is created by Initiate Project or Phase in addition to the project charter?

- **A.** Risk register
- **B.** Assumption log
- **C.** Cost baseline
- **D.** Stakeholder engagement plan

35. A project manager compares actual progress with the plan and reports that the project is 12% behind schedule. This is an example of:

- **A.** A deliverable
- **B.** An enterprise environmental factor
- **C.** Work performance information
- **D.** Work performance data

36. Which development approach is usually most appropriate when the cost of change is high and requirements are stable and well understood?

- **A.** Predictive
- **B.** Iterative experimentation
- **C.** Adaptive
- **D.** Kanban flow

37. Which of the following is NOT a component of the project management plan?

- **A.** Issue log
- **B.** Scope baseline
- **C.** Change management plan
- **D.** Risk management plan

38. In an adaptive project, when do the five focus areas occur?

- **A.** Once each, in strict sequence, across the whole of the project
- **B.** Within each iteration, repeatedly throughout the project
- **C.** Only during the first iteration
- **D.** Only during release planning

39. Which two processes from the Sixth Edition were merged into Perform Risk Analysis?

- **A.** Plan Risk Management and Identify Risks, to simplify early risk planning
- **B.** Perform Qualitative Risk Analysis and Perform Quantitative Risk Analysis
- **C.** Identify Risks and Perform Qualitative Risk Analysis, to speed up risk assessment

- **D.** Plan Risk Responses and Implement Risk Responses, to link planning and action

40. A company releases a new feature to all customers at the end of every month. Which delivery cadence is this?

- **A.** Ad hoc delivery
- **B.** Continuous delivery
- **C.** Single delivery
- **D.** Periodic delivery

41. Which process consolidates subsidiary plans into a cohesive project management plan?

- **A.** Assess and Implement Changes
- **B.** Integrate and Align Project Plans
- **C.** Initiate Project or Phase
- **D.** Plan Scope Management

42. An organisation is unsure whether a project is clear, complicated, complex or chaotic. According to Cynefin, what is this state called?

- **A.** Confusion (disorder)
- **B.** Chaotic
- **C.** Complex
- **D.** Clear

43. Match each process with its focus area.

(Matching: pair every numbered item with one lettered option.)

Item	Option
1. Identify Stakeholders	**A.** Planning
2. Develop Budget	**B.** Monitoring and Controlling
3. Lead the Team	**C.** Executing
4. Monitor Communications	**D.** Initiating

44. Which type of change request is a modification to a non-conforming product component?

- **A.** Preventive action
- **B.** Defect repair
- **C.** Corrective action
- **D.** Update

45. What is the main reason for holding a kickoff meeting?

- **A.** To obtain formal customer acceptance of the deliverables produced so far
- **B.** To align everyone on objectives, roles and the plan as the team moves into execution
- **C.** To obtain the sponsor's signature on the project charter and formally authorise the project
- **D.** To review lessons learned from the previous phase and close open procurements

Answers — Part 2 (Life Cycles, Focus Areas and Processes)

Q26 — Answer: C

Correct: C. Iterative development with a single delivery

Repeated refinement through prototypes is iterative; delivering everything at once is a single delivery.

Q27 — Answer: C

Correct: C. Planning

Planning has 19 of the 40 processes, Monitoring and Controlling 10, Executing 8, Initiating 2 and Closing 1.

Q28 — Answer: D

Correct: D. Implement Risk Responses

Implement Risk Responses is the only executing process in the Risk domain. Analysis and response planning are planning processes; Monitor Risks is monitoring and controlling.

Q29 — Answer: C

Correct: C. Validate Scope

Validate Scope (formal acceptance) and Monitor and Control Scope are the two monitoring and controlling processes of the Scope domain.

Q30 — Answer: A

Correct: A. Governance

Plan Sourcing Strategy is a Governance planning process, reflecting that sourcing decisions are strategic. Acquisition and control of resources are in the Resources domain.

Q31 — Answer: C

Correct: C. Manage Project Execution

Manage Project Execution in the Governance domain is the renamed Direct and Manage Project Work.

Q32 — Answer: A

Correct: A. Continue with modifications

Phase gate decisions can be to continue, continue with modifications, repeat the phase or end the project.

Q33 — Answer: B, E

Correct: B. The issue log can be updated without a formal change request / E. Changes to the schedule baseline require formal change control

Project documents such as the issue log are updated as needed; baselines and the plan require change control. Registers are not baseline components.

Q34 — Answer: B

Correct: B. Assumption log

The charter and the assumption log are the key outputs of initiating the project or phase. The risk register is created in Identify Risks.

Q35 — Answer: C

Correct: C. Work performance information

Data that has been analysed against the plan becomes work performance information.

Q36 — Answer: A

Correct: A. Predictive

Predictive approaches suit stable requirements and expensive change, such as construction.

Q37 — Answer: A

Correct: A. Issue log

The issue log is a project document. Baselines, subsidiary plans and the change management plan are plan components.

Q38 — Answer: B

Correct: B. Within each iteration, repeatedly throughout the project

Planning, executing, monitoring and controlling, and closing occur in every iteration; initiating recurs with new releases or phases.

Q39 — Answer: B

Correct: B. Perform Qualitative Risk Analysis and Perform Quantitative Risk Analysis

The two analysis processes were combined into a single process covering both qualitative and quantitative techniques.

Q40 — Answer: D

Correct: D. Periodic delivery

Fixed, regular releases are periodic. Continuous delivery releases whenever features are ready.

Q41 — Answer: B

Correct: B. Integrate and Align Project Plans

Integrate and Align Project Plans is the Eighth Edition name for developing the project management plan.

Q42 — Answer: A

Correct: A. Confusion (disorder)

Cynefin's central domain, confusion, applies when it is not yet known which domain the situation belongs to.

Q43 — Answer: 1→D, 2→A, 3→C, 4→B

Correct pairs: 1. Identify Stakeholders → Initiating; 2. Develop Budget → Planning; 3. Lead the Team → Executing; 4. Monitor Communications → Monitoring and Controlling

Identify Stakeholders is initiating, Develop Budget planning, Lead the Team executing, and Monitor Communications monitoring and controlling.

Q44 — Answer: B

Correct: B. Defect repair

Defect repair modifies a non-conforming product. Corrective and preventive actions address performance; updates change controlled documents.

Q45 — Answer: B

Correct: B. To align everyone on objectives, roles and the plan as the team moves into execution

The kickoff aligns stakeholders and the team on objectives, roles and the plan, typically at the end of planning or the start of execution.

Part 3 — Governance

46. The governance framework states that the project manager may approve changes up to $10,000; above that, the change control board decides. A change estimated at $25,000 is requested. What should the project manager do?

- **A.** Approve it, since the project manager is accountable for delivering the project outcomes
- **B.** Reject it, since changes above the project manager's threshold cannot be accommodated
- **C.** Analyse the impact and submit the request, with the analysis, to the change control board
- **D.** Split it into several smaller changes that each fall within the project manager's approval limit

47. A change request was approved by the change control board. What should happen next?

- **A.** Implement the change right away and update the documentation at the next phase gate
- **B.** Hold the change until the next release so that the current baseline is not disturbed
- **C.** Update the affected plans, baselines and change log, communicate it and implement it
- **D.** Ask the sponsor to confirm the decision before informing the team about the change

48. A team member discovers that the version of a design drawing used on site is not the latest approved version. Which governance mechanism should have prevented this?

- **A.** Stakeholder engagement planning
- **B.** Earned value management
- **C.** Configuration management
- **D.** Resource levelling

49. Which statement about the sponsor's role in governance is most accurate?

- **A.** Approves every work package and activity before the team is allowed to start working on it
- **B.** Champions the project, provides resources, decides beyond the PM's authority and owns benefits
- **C.** Develops the detailed schedule and cost estimates on behalf of the project manager and team
- **D.** Manages the team's daily work and assigns tasks so that the project stays on schedule and budget

50. A project manager wants to capture tacit knowledge from senior engineers before they retire during the project. Which approach is most effective?

- **A.** Ask the senior engineers to complete a detailed survey about all of their experience before leaving
- **B.** Copy the senior engineers' email archives and documents to the shared project repository
- **C.** Invite the senior engineers to the final lessons learned meeting at the end of the project
- **D.** Pair junior engineers with the senior engineers and run job-shadowing and storytelling sessions

51. During a quality audit, the auditor finds that the team is not following the agreed peer-review procedure. Which process does this audit belong to?

- **A.** Close Project or Phase
- **B.** Validate Scope
- **C.** Manage Quality Assurance
- **D.** Monitor and Control Scope

52. The project's phase gate review is scheduled, but the team is confident the project is on track. What is the main purpose of still holding the gate review?

- **A.** To confirm the project still aligns with the business case and authorise the next phase
- **B.** To re-estimate every remaining activity so that the whole schedule baseline can be refreshed
- **C.** To approve the team's planned leave and confirm resource calendars for the next phase
- **D.** To close the procurement contracts that supported the work of the completed phase

53. What is the purpose of the change log?

- **A.** To record identified risks together with their planned responses
- **B.** To track change requests, their status and the decisions made
- **C.** To capture knowledge gained so that future projects can benefit
- **D.** To record how each stakeholder has been classified and prioritised

54. A software vendor's project manager agrees verbally with the buyer's project manager to add three reports at no cost. What is the main problem?

- **A.** Contract changes must go through contract change control and be authorised in writing
- **B.** Verbal agreements between project managers are illegal in most jurisdictions
- **C.** Reports cannot be part of a contract, so they must be procured separately
- **D.** There is no problem, because the seller has agreed to add the reports to the contract at no cost

55. Which contract type is most suitable for a five-year supply agreement in which the cost of steel may fluctuate significantly?

- **A.** Cost plus award fee
- **B.** Firm fixed price
- **C.** Time and materials
- **D.** Fixed price with economic price adjustment

56. A buyer cannot define the scope of a research project. Which contract type places most cost risk on the buyer while still motivating the seller through a fee tied to performance?

- **A.** Fixed price with economic price adjustment
- **B.** Cost plus incentive fee
- **C.** Firm fixed price
- **D.** Fixed price incentive fee

57. Which document describes the items to be procured in sufficient detail for prospective sellers to determine whether they can provide them?

- **A.** Seller's proposal
- **B.** Bid evaluation report
- **C.** Source selection criteria
- **D.** Procurement statement of work

58. A project is cancelled after 40% of the work is done. Which activity is still required?

- **A.** Completing the remaining work so that all deliverables are available for the record
- **B.** Nothing, because closure activities apply only to projects that finish successfully
- **C.** Archiving the charter only, since other documents are no longer relevant to anyone

- **D.** Documenting the reasons and status, closing contracts and capturing lessons learned

59. Which two practices reflect governance tailored for adaptive delivery? (Choose two.)

- **A.** Requiring CCB approval for every user story change
- **B.** Using guardrails such as budget limits and architecture standards within which the team decides
- **C.** Freezing all requirements before the first sprint
- **D.** Funding the product team incrementally based on demonstrated value
- **E.** Holding weekly 4-hour status meetings with all developers

60. A project manager observes that the team keeps working on change requests that were never approved because "the customer asked nicely". What should the project manager do?

- **A.** Remove the customer's access to the team for the rest of the project to stop new requests
- **B.** Stop the project until the customer agrees in writing never to request changes again
- **C.** Reinforce the change control process with the team and customer, and assess the work done
- **D.** Accept the practice, because customer satisfaction is the most important success criterion

61. Which output of Manage Project Execution provides the raw data used by the monitoring and controlling processes?

- **A.** The project charter
- **B.** Work performance data
- **C.** Accepted deliverables
- **D.** Work performance reports

62. A project manager receives a change request from the client. After impact analysis, the change would delay the project by three weeks and add $40,000. The client insists the change is essential. What is the best next step?

- **A.** Refuse the change, since it increases cost and the cost baseline has already been approved
- **B.** Ask the team to absorb the work through overtime so that the baseline does not need to change
- **C.** Present the impact analysis and options to the change control board or authorised decision-maker
- **D.** Implement the change, since the client considers it essential and will accept the resulting delay later

63. Which artifact gives the project manager the authority to apply organisational resources?

- **A.** Business case
- **B.** Project charter
- **C.** Project management plan
- **D.** Team charter

64. During closure, the project manager realises that some lessons were never captured. Which practice would have prevented this?

- **A.** Holding a longer lessons learned workshop at the end of the project with all members
- **B.** Asking the sponsor to write the lessons learned, since the sponsor has the full overview
- **C.** Capturing lessons learned only on projects that encountered serious problems
- **D.** Recording lessons learned throughout the project in the lessons learned register

65. Which source selection method evaluates proposals on a combination of technical quality and price using weighted criteria?

- **A.** Qualifications only
- **B.** Quality and cost-based selection
- **C.** Sole source
- **D.** Least cost

66. What is the main difference between corrective and preventive actions?

- **A.** Corrective actions realign current performance; preventive actions keep future performance aligned
- **B.** Corrective actions repair defective products; preventive actions update the project documents
- **C.** Corrective actions are used only in predictive projects, and preventive actions only in agile projects
- **D.** Corrective actions are approved by the sponsor; preventive actions are approved by the team

67. A project in a regulated industry needs a decision about a design change that affects safety. The project manager's authority covers only cost changes under $5,000. The change costs $2,000. What should the project manager consider?

- **A.** Approving it at once, since the cost is well below the project manager's own approval threshold
- **B.** Implementing it immediately, since safety changes are urgent and bypass governance
- **C.** Whether governance requires safety-related changes to be approved by a specific authority
- **D.** Asking the team to vote on the change, since they understand the design best

68. Which meeting marks the end of planning and the start of execution in a predictive project?

- **A.** Close-out meeting
- **B.** Bidder conference
- **C.** Kickoff meeting
- **D.** Sprint retrospective

69. The steering committee asks for a single integrated view of progress for a hybrid project with predictive hardware work and agile software work. What should the project manager provide?

- **A.** An integrated report combining milestones, release burnup, key risks and forecasts
- **B.** A list of individual developers' velocity so the committee can compare productivity
- **C.** Only the hardware Gantt chart, since the software scope is flexible and changes often
- **D.** Two separate reports, one for hardware and one for software, without a combined view

70. What is the PMO's typical role in a controlling PMO model?

- **A.** Approves the sponsor's budget and benefits targets for each project
- **B.** Offers optional templates and training without any compliance requirements
- **C.** Provides support and requires compliance with frameworks, templates and reviews
- **D.** Directly manages all of the projects, with every project manager reporting to the PMO

Answers — Part 3 (Governance)

Q46 — Answer: C

Correct: C. Analyse the impact and submit the request, with the analysis, to the change control board

Authority thresholds define who decides. The PM prepares the impact analysis and submits it to the CCB. Splitting the change to avoid governance is unethical.

Q47 — Answer: C

Correct: C. Update the affected plans, baselines and change log, communicate it and implement it

Approved changes are reflected in plans and baselines, logged, communicated and then implemented in Manage Project Execution.

Q48 — Answer: C

Correct: C. Configuration management

Configuration management controls versions of product specifications and documents through identification, status accounting and verification.

Q49 — Answer: B

Correct: B. Champions the project, provides resources, decides beyond the PM's authority and owns benefits

Sponsors provide direction, funding and support, and resolve escalated issues; they do not manage day-to-day work.

Q50 — Answer: D

Correct: D. Pair junior engineers with the senior engineers and run job-shadowing and storytelling sessions

Tacit knowledge transfers best through interaction such as pairing, shadowing and storytelling, and should be captured while people are still available.

Q51 — Answer: C

Correct: C. Manage Quality Assurance

Audits of process compliance are part of quality assurance in the Governance domain.

Q52 — Answer: A

Correct: A. To confirm the project still aligns with the business case and authorise the next phase

Phase gates are governance decision points that confirm continued justification and authorise continuation, even when performance is good.

Q53 — Answer: B

Correct: B. To track change requests, their status and the decisions made

The change log tracks change requests submitted, their impact assessments, decisions and status.

Q54 — Answer: A

Correct: A. Contract changes must go through contract change control and be authorised in writing

Contract changes require formal documentation and approval by authorised parties. Informal changes create disputes and uncontrolled scope.

Q55 — Answer: D

Correct: D. Fixed price with economic price adjustment

FP-EPA contracts include adjustments for defined economic changes, such as commodity prices, over long periods.

Q56 — Answer: B

Correct: B. Cost plus incentive fee

Cost-reimbursable contracts suit uncertain scope; CPIF links the fee to performance against targets.

Q57 — Answer: D

Correct: D. Procurement statement of work

The procurement SOW describes the item or service to be procured; source selection criteria explain how proposals will be evaluated.

Q58 — Answer: D

Correct: D. Documenting the reasons and status, closing contracts and capturing lessons learned

Projects closed early still need closure activities to settle obligations, transfer partial work and preserve knowledge.

Q59 — Answer: B, D

Correct: B. Using guardrails such as budget limits and architecture standards within which the team decides / D. Funding the product team incrementally based on demonstrated value

Lightweight governance uses incremental funding and guardrails. Detailed CCB control of stories and frozen requirements contradict adaptive delivery.

Q60 — Answer: C

Correct: C. Reinforce the change control process with the team and customer, and assess the work done

Unapproved changes cause scope creep. The PM coaches the team and the customer on the agreed process and assesses the impact of work already done.

Q61 — Answer: B

Correct: B. Work performance data

Executing processes produce work performance data, which control processes analyse into information and then reports.

Q62 — Answer: C

Correct: C. Present the impact analysis and options to the change control board or authorised decision-maker

The decision belongs to those with authority, who need the analysis and options. The PM neither refuses nor implements unilaterally.

Q63 — Answer: B

Correct: B. Project charter

The charter formally authorises the project and the PM's authority.

Q64 — Answer: D

Correct: D. Recording lessons learned throughout the project in the lessons learned register

Lessons should be recorded continuously, when memories are fresh and while they can still improve the current project.

Q65 — Answer: B

Correct: B. Quality and cost-based selection

Quality and cost-based selection weighs both technical merit and price. Least cost suits standard items; sole source has one supplier.

Q66 — Answer: A

Correct: A. Corrective actions realign current performance; preventive actions keep future performance aligned

Corrective action addresses deviations that have occurred; preventive action reduces the chance of future deviations.

Q67 — Answer: C

Correct: C. Whether governance requires safety-related changes to be approved by a specific authority

Governance can define thresholds by impact type (safety, compliance), not only cost. The PM checks the decision rights before acting.

Q68 — Answer: C

Correct: C. Kickoff meeting

The kickoff meeting launches execution by aligning stakeholders on objectives, roles and the plan.

Q69 — Answer: A

Correct: A. An integrated report combining milestones, release burnup, key risks and forecasts

Hybrid governance benefits from one coherent story combining predictive and adaptive measures, without exposing inappropriate individual metrics.

Q70 — Answer: C

Correct: C. Provides support and requires compliance with frameworks, templates and reviews

Controlling PMOs provide support and require compliance; directive PMOs manage projects; supportive PMOs consult.

Part 4 — Scope and Quality

71. Which technique is most appropriate when users have difficulty describing how they currently perform a complex process?

- **A.** Observation (job shadowing)
- **B.** Benchmarking
- **C.** Questionnaires
- **D.** Voting

72. A requirements workshop brings together users, developers and testers from different departments to define requirements quickly and resolve conflicts. In software development, this is often called:

- **A.** A retrospective
- **B.** Joint application design (JAD)
- **C.** Quality function deployment only
- **D.** A bidder conference

73. Which statement about the WBS is correct?

- **A.** It lists all project activities in the chronological order in which they will be performed
- **B.** It describes only the work that external contractors will perform under the contract
- **C.** It is a deliverable-oriented hierarchical decomposition of the total scope of work
- **D.** It shows the reporting relationships between the project team and functional departments

74. A control account manager compares the budget, schedule and earned value for a group of work packages. What is a control account?

- **A.** The lowest level of the WBS, where activities are defined and sequenced
- **B.** A summary of the risks that could affect a group of related work packages
- **C.** A dedicated bank account that is used to receive and disburse all of the project's funds
- **D.** A management point where scope, budget and schedule are integrated and measured

75. The customer reviews the completed training materials and signs an acceptance form. Which output has been produced?

- **A.** Accepted deliverables
- **B.** Quality control measurements
- **C.** Work performance data
- **D.** Verified deliverables

76. Which document should be checked to determine whether a requested feature is out of scope?

- **A.** The project scope statement, especially the exclusions
- **B.** The resource calendar for the team members involved
- **C.** The communications management plan and its distribution list
- **D.** The team charter and its decision-making guidelines

77. A product owner uses the categories Must have, Should have, Could have and Won't have this time. Which technique is this?

- **A.** Kano analysis
- **B.** MoSCoW prioritisation
- **C.** Pareto analysis
- **D.** Monte Carlo simulation

78. Which two statements correctly describe the definition of done? (Choose two.)

- **A.** It is a shared understanding of the quality criteria for an increment
- **B.** It applies to every backlog item the team completes
- **C.** It replaces acceptance criteria entirely
- **D.** It is only reviewed at project closure
- **E.** It is written separately by the customer for each user story

79. During testing, the team finds that a deliverable does not meet the specified tolerance. Which process detects this?

- **A.** Monitor and Control Scope (quality control)
- **B.** Elicit and Analyze Requirements (analysis)
- **C.** Manage Quality Assurance (process audit)
- **D.** Validate Scope (customer acceptance review)

80. A manufacturing project spends money on inspecting incoming materials. Which cost of quality category is this?

- **A.** Prevention
- **B.** External failure
- **C.** Appraisal
- **D.** Internal failure

81. A process's control chart shows one point above the upper control limit. What does this mean?

- **A.** The process is out of control and the special cause should be investigated
- **B.** The control limits were calculated wrongly and must be recalculated at once
- **C.** The process is in control because only a single point is outside the limit
- **D.** The specification limits should be adjusted to include the new measurement

82. A Pareto analysis shows that 3 of 15 defect types cause 78% of customer complaints. What should the team do?

- **A.** Focus improvement efforts on those three defect types first
- **B.** Address all fifteen defect types equally so that none is neglected
- **C.** Collect another year of data before deciding where to focus improvements
- **D.** Start with the least frequent defect types because they are quickest to fix

83. Which quality tool is used to tally occurrences of defects by type during inspections?

- **A.** Scatter diagram
- **B.** Check sheet
- **C.** Flowchart

- **D.** Control chart

84. What does the requirements traceability matrix help a project manager do when a stakeholder requests a change to a requirement?

- **A.** Determine which team members are available to implement the change
- **B.** Calculate the cost performance index of the affected work packages
- **C.** Identify the deliverables, designs and tests affected by the change
- **D.** Decide which contract type should be used for the changed requirement

85. A team wants to verify whether training hours correlate with lower defect rates. Which tool fits best?

- **A.** Pareto chart
- **B.** Histogram
- **C.** Scatter diagram
- **D.** Fishbone diagram

86. Which statement about Validate Scope is correct?

- **A.** It is performed only once, at the very end of the project, during the final closing meeting
- **B.** It is another name for quality control and is carried out by the testing team
- **C.** It is performed by the project team alone, without involving the customer
- **D.** It is performed periodically with the customer or sponsor, usually after quality control

87. A team member wants to add an extra report "because it would be nice to have". The scope baseline does not include it. What is the best response?

- **A.** Let the team member add it quietly, since small extras help to impress the customer
- **B.** Add the report and remove a required report so that the total effort stays the same
- **C.** Approve it, since a more complete set of reports always improves product quality
- **D.** Explain that it would be gold plating and suggest raising it with the product owner

88. Which document describes, for each WBS element, the work, responsible organisation, milestones, cost estimates and acceptance criteria?

- **A.** Quality report
- **B.** WBS dictionary
- **C.** Requirements documentation
- **D.** Project charter

89. In the Kano model, a hotel guest expects hot water in the shower and is very dissatisfied without it, but having it does not increase satisfaction. This feature is:

- **A.** An indifferent feature
- **B.** An excitement (delighter) feature
- **C.** A performance (linear) feature
- **D.** A basic (must-be) quality

90. A product owner asks the team to split a large epic. Which characteristic of good user stories does this mainly support?

- **A.** Small enough to be completed within one iteration
- **B.** Dependent on other stories so they are delivered together

- **C.** Fixed and non-negotiable once written by the product owner
- **D.** Valuable mainly to the developers who will implement it

91. Which practice best reflects "prevention over inspection"?

- **A.** Doubling the number of final inspectors so that more defects are caught before shipping
- **B.** Relaxing the specification limits so fewer units are classified as defective
- **C.** Letting customers report defects and then fixing them quickly through support
- **D.** Training the team and using checklists and automated tests before defects occur

92. Match each quality concept with its description.

(Matching: pair every numbered item with one lettered option.)

Item	Option
1. Grade	**A.** Measured value is close to the true value
2. Precision	**B.** Repeated measurements are closely clustered
3. Accuracy	**C.** Acceptable range of variation for a quality metric
4. Tolerance	**D.** Category assigned to products with the same use but different characteristics

93. In the Eighth Edition, which process includes planning how quality will be achieved, such as defining quality metrics?

- **A.** Define Scope
- **B.** Develop Budget
- **C.** Close Project or Phase
- **D.** Plan Communications Management

94. A requirement states: "Customer data must be migrated from the legacy system before go-live." Which requirement type is this?

- **A.** Non-functional requirement
- **B.** Project requirement only
- **C.** Business requirement
- **D.** Transition and readiness requirement

95. A product's process is in statistical control, but many units still fall outside the customer's specifications. What does this indicate?

- **A.** The customer's specifications are unrealistic and should be changed to match the process
- **B.** The process is stable but not capable of meeting the specification, so it must be improved
- **C.** The process is out of statistical control and its special causes must be investigated
- **D.** The control limits should be widened so that they match the customer's specification limits

Answers — Part 4 (Scope and Quality)

Q71 — Answer: A

Correct: A. Observation (job shadowing)

Observation reveals hidden or tacit requirements when people struggle to articulate how they work.

Q72 — Answer: B

Correct: B. Joint application design (JAD)

JAD sessions are facilitated workshops for software requirements. QFD (voice of the customer) is a similar technique used in manufacturing.

Q73 — Answer: C

Correct: C. It is a deliverable-oriented hierarchical decomposition of the total scope of work

The WBS decomposes scope into deliverables and work packages. Sequencing is done in the schedule; reporting lines are shown in an OBS.

Q74 — Answer: D

Correct: D. A management point where scope, budget and schedule are integrated and measured

Control accounts integrate scope, budget and schedule and contain one or more work packages.

Q75 — Answer: A

Correct: A. Accepted deliverables

Formal sign-off by the customer produces accepted deliverables in Validate Scope.

Q76 — Answer: A

Correct: A. The project scope statement, especially the exclusions

The scope statement defines deliverables and exclusions; it clarifies whether the request is within scope.

Q77 — Answer: B

Correct: B. MoSCoW prioritisation

MoSCoW classifies requirements by priority for a timebox or release.

Q78 — Answer: A, B

Correct: A. It is a shared understanding of the quality criteria for an increment / B. It applies to every backlog item the team completes

The DoD is a team-wide quality standard applied to all work. Acceptance criteria remain specific to each story.

Q79 — Answer: A

Correct: A. Monitor and Control Scope (quality control)

Inspection and testing of deliverables against requirements is quality control, part of Monitor and Control Scope in the Eighth Edition.

Q80 — Answer: C

Correct: C. Appraisal

Inspection and testing are appraisal costs incurred to assess quality.

Q81 — Answer: A

Correct: A. The process is out of control and the special cause should be investigated

A single point outside the control limits indicates a special cause of variation that should be investigated.

Q82 — Answer: A

Correct: A. Focus improvement efforts on those three defect types first

The Pareto principle directs effort to the vital few causes that create most of the problems.

Q83 — Answer: B

Correct: B. Check sheet

Check sheets organise data collection by category, and their results are often used to build Pareto diagrams.

Q84 — Answer: C

Correct: C. Identify the deliverables, designs and tests affected by the change

Traceability links requirements forward to deliverables and tests, supporting impact analysis.

Q85 — Answer: C

Correct: C. Scatter diagram

Scatter diagrams plot two variables to identify correlation.

Q86 — Answer: D

Correct: D. It is performed periodically with the customer or sponsor, usually after quality control

Validate Scope is repeated for deliverables over time; it obtains formal acceptance after deliverables have been verified.

Q87 — Answer: D

Correct: D. Explain that it would be gold plating and suggest raising it with the product owner

Gold plating adds cost and risk without agreed value. If the idea has merit, it goes through the appropriate channel.

Q88 — Answer: B

Correct: B. WBS dictionary

The WBS dictionary provides detailed information for each WBS element and is part of the scope baseline.

Q89 — Answer: D

Correct: D. A basic (must-be) quality

Basic qualities cause dissatisfaction when absent but no extra satisfaction when present.

Q90 — Answer: A

Correct: A. Small enough to be completed within one iteration

INVEST stories are Small enough to fit in an iteration, as well as Independent, Negotiable, Valuable, Estimable and Testable.

Q91 — Answer: D

Correct: D. Training the team and using checklists and automated tests before defects occur

Prevention reduces the chance of defects; it is cheaper than finding them through inspection or by customers.

Q92 — Answer: 1→D, 2→B, 3→A, 4→C

Correct pairs: 1. Grade → Category assigned to products with the same use but different characteristics; 2. Precision → Repeated measurements are closely clustered; 3. Accuracy → Measured value is close to the true value; 4. Tolerance → Acceptable range of variation for a quality metric

Grade differs from quality; precision is consistency; accuracy is correctness; tolerance is the allowed variation.

Q93 — Answer: A

Correct: A. Define Scope

Quality planning was incorporated into Define Scope.

Q94 — Answer: D

Correct: D. Transition and readiness requirement

Transition requirements describe temporary capabilities needed to move from the current to the future state, such as data conversion and training.

Q95 — Answer: B

Correct: B. The process is stable but not capable of meeting the specification, so it must be improved

Control limits describe the process; specification limits describe customer requirements. A stable process can still fail to meet specifications.

Part 5 — Schedule

96. Which document defines the scheduling methodology, units of measure, control thresholds and rules of performance measurement?

- **A.** Activity attributes
- **B.** Schedule baseline
- **C.** Milestone list
- **D.** Schedule management plan

97. What is the duration of a milestone?

- **A.** It depends on the critical path
- **B.** Zero
- **C.** One day
- **D.** The length of the longest activity

98. An activity requires concrete to cure for 7 days before the next activity can start. How is this represented?

- **A.** Seven days of free float on the curing activity
- **B.** A start-to-finish relationship with a 7-day duration
- **C.** A lead of 7 days on the finish-to-start relationship
- **D.** A lag of 7 days on the finish-to-start relationship

99. A team decides to start writing the user manual 5 days before the software is finished. This is an example of:

- **A.** A lag
- **B.** A mandatory dependency
- **C.** A lead
- **D.** Resource levelling

100. Network: A (2 days, start), B (4, after A), C (6, after A), D (3, after B), E (2, after C and D). What is the critical path?

- **A.** A-C (8 days)
- **B.** A-B-D (9 days)
- **C.** A-B-D-E (11 days)
- **D.** A-C-E (10 days)

101. In the same network (A 2, B 4 after A, C 6 after A, D 3 after B, E 2 after C and D), what is the total float of activity C?

- **A.** 1 day
- **B.** 3 days
- **C.** 0 days
- **D.** 2 days

102. Two critical-path activities can be shortened. Activity X: 2 days saved for $6,000 total. Activity Y: 2 days saved for $4,000 total. The project needs to save 2 days at the least cost. What should be crashed?

- **A.** A non-critical activity
- **B.** Both X and Y
- **C.** Activity Y
- **D.** Activity X

103. Which risk is most associated with fast tracking?

- **A.** Higher direct costs, because overtime and extra staff must be paid for the compressed work
- **B.** Lower resource availability for other projects in the portfolio
- **C.** A longer overall duration, because activities wait for each other
- **D.** Rework, because activities start before their predecessors are complete

104. A specialist is assigned to two activities scheduled at the same time. The project manager delays one activity until the specialist is free, extending the project end date. Which technique was used?

- **A.** Resource levelling
- **B.** Resource smoothing
- **C.** Fast tracking
- **D.** Crashing

105. Which estimating technique uses historical data and a statistical relationship, such as cost per square metre?

- **A.** Expert judgment only
- **B.** Bottom-up
- **C.** Parametric
- **D.** Analogous

106. Which estimating technique is generally the most accurate but also the most time-consuming?

- **A.** Rough order of magnitude
- **B.** Top-down
- **C.** Analogous
- **D.** Bottom-up

107. Activity estimates: O = 8, M = 11, P = 20 days. What is the triangular (simple average) estimate in days?

(Fill in the blank.)

Answer: _____

108. A release burnup chart shows the total-scope line rising every sprint while the completed-work line rises steadily. What is the most likely conclusion?

- **A.** The project is ahead of schedule because more work is being added
- **B.** Scope is growing, which may delay the release unless scope is managed
- **C.** The team's velocity is decreasing and the team needs more coaching
- **D.** The team has stopped delivering any completed work during the most recent sprints

109. Which critical chain element protects the project end date from variation along the critical chain?

- **A.** Project buffer
- **B.** Free float
- **C.** Resource buffer
- **D.** Feeding buffer

110. A Kanban team wants to reduce the time items take to move from "in progress" to "done". What is the most direct lever according to Little's Law?

- **A.** Estimate items in hours instead of points
- **B.** Increase the number of columns on the board
- **C.** Reduce work in progress
- **D.** Start more work at once

111. Which chart is most appropriate to show senior executives only the key dates of a project?

- **A.** Activity list
- **B.** Resource histogram
- **C.** Milestone chart
- **D.** Detailed network diagram

112. A project manager plans the next two months in detail and leaves later work as planning packages. This approach is called:

- **A.** Crashing
- **B.** Fast tracking
- **C.** Critical chain
- **D.** Rolling wave planning

113. Which two statements about free float are correct? (Choose two.)

- **A.** It is always equal to total float
- **B.** It applies only to critical-path activities
- **C.** It is the time an activity can be delayed without delaying the early start of any successor
- **D.** It can never be greater than the activity's total float
- **E.** It is the time the project can be delayed without penalty

114. The team's velocity for the last four sprints was 18, 22, 20 and 20. The remaining backlog for the release is 120 points. How many sprints should be planned?

- **A.** 5
- **B.** 6
- **C.** 7
- **D.** 8

115. A non-critical activity with 8 days of total float is delayed by 3 days. What is the effect on the project end date?

- **A.** None, as long as the delay stays within the float and nothing else changes
- **B.** The project end date moves by 3 days, as with any delayed activity
- **C.** The project end date moves by 8 days, the full amount of the float

- **D.** The activity becomes critical immediately, and the project end date slips by 3 days

Answers — Part 5 (Schedule)

Q96 — Answer: D

Correct: D. Schedule management plan

The schedule management plan, created in Plan Schedule Management, sets the rules for developing and controlling the schedule.

Q97 — Answer: B

Correct: B. Zero

Milestones mark significant points or events and have zero duration.

Q98 — Answer: D

Correct: D. A lag of 7 days on the finish-to-start relationship

A lag is a required delay between the predecessor and successor.

Q99 — Answer: C

Correct: C. A lead

A lead allows the successor to start before the predecessor finishes.

Q100 — Answer: C

Correct: C. A-B-D-E (11 days)

Path A-B-D-E = 2 + 4 + 3 + 2 = 11 days; path A-C-E = 2 + 6 + 2 = 10 days. The longest path, A-B-D-E, is critical.

Q101 — Answer: A

Correct: A. 1 day

C can start at day 2 and finish at day 8; E starts at day 9 (after D). C's late finish is 9, so its float is 9 − 8 = 1 day.

Q102 — Answer: C

Correct: C. Activity Y

Crash the critical activity with the lowest cost per day saved: Y costs $2,000 per day versus $3,000 for X.

Q103 — Answer: D

Correct: D. Rework, because activities start before their predecessors are complete

Fast tracking overlaps activities, increasing the risk of rework. Crashing increases cost.

Q104 — Answer: A

Correct: A. Resource levelling

Resource levelling resolves over-allocation even if the critical path and end date change. Smoothing stays within float.

Q105 — Answer: C

Correct: C. Parametric

Parametric estimating multiplies a quantity by a historical unit rate or uses other statistical relationships.

Q106 — Answer: D

Correct: D. Bottom-up

Bottom-up estimates each component in detail and aggregates the results.

Q107 — Answer: 13

Triangular estimate = (O + M + P) / 3 = (8 + 11 + 20) / 3 = 39 / 3 = 13 days.

Q108 — Answer: B

Correct: B. Scope is growing, which may delay the release unless scope is managed

The rising scope line reveals scope growth; the completed line shows the team continues to deliver.

Q109 — Answer: A

Correct: A. Project buffer

The project buffer is placed at the end of the critical chain. Feeding buffers protect where non-critical chains join it.

Q110 — Answer: C

Correct: C. Reduce work in progress

Cycle time equals WIP divided by throughput; reducing WIP reduces cycle time when throughput is stable.

Q111 — Answer: C

Correct: C. Milestone chart

Milestone charts show only major events and are suited for executive audiences.

Q112 — Answer: D

Correct: D. Rolling wave planning

Rolling wave planning details near-term work and elaborates later work progressively.

Q113 — Answer: C, D

Correct: C. It is the time an activity can be delayed without delaying the early start of any successor / D. It can never be greater than the activity's total float

Free float concerns successors' early starts and cannot exceed total float, which concerns the project end date.

Q114 — Answer: B

Correct: B. 6

Average velocity = 80 / 4 = 20 points. 120 / 20 = 6 sprints.

Q115 — Answer: A

Correct: A. None, as long as the delay stays within the float and nothing else changes

A delay within total float does not affect the end date, although the remaining float is reduced to 5 days.

Part 6 — Finance

116. Which component of the financial management plan specifies the acceptable range used to determine realistic cost estimates, such as ±10%?

- **A.** Units of measure
- **B.** Level of accuracy
- **C.** Funding sources
- **D.** Level of precision

117. A project's indirect costs include:

- **A.** Steel purchased specifically for the project's main structure
- **B.** Equipment rented only for the project
- **C.** Salaries of the dedicated project team
- **D.** A share of the company's office rent and utilities

118. Which statement about the cost baseline and project budget is correct?

- **A.** Cost baseline = project budget + management reserve
- **B.** Cost baseline excludes contingency reserves
- **C.** Project budget excludes the cost baseline
- **D.** Project budget = cost baseline + management reserve

119. A project is 50% complete according to the schedule plan, but only 40% of the work is actually done. BAC is $400,000 and $180,000 has been spent. What is the CPI?

- **A.** 0.80
- **B.** 0.89
- **C.** 0.90
- **D.** 1.11

120. Using the same data (BAC $400,000, planned 50%, actual 40% complete, AC $180,000), what is the SPI?

- **A.** 0.80
- **B.** 0.89
- **C.** 0.90
- **D.** 1.25

121. CPI = 0.9 and SPI = 0.8. BAC = $1,000,000, EV = $400,000, AC = $444,444. If both cost and schedule performance are expected to affect the remaining work, what is the EAC (rounded)?

- **A.** About $1,111,100
- **B.** About $1,000,000
- **C.** About $1,044,400
- **D.** About $1,277,800

122. What does a VAC of −$50,000 indicate?

- **A.** The project is $50,000 under budget today

- **B.** The project is $50,000 behind schedule
- **C.** The project is expected to finish $50,000 over budget
- **D.** The project has $50,000 of management reserve left to spend

123. A project manager discovers that the original cost estimates were based on wrong assumptions about the technology. Which EAC method is most appropriate?

- **A.** AC plus (BAC − EV)
- **B.** BAC
- **C.** AC plus a new bottom-up estimate to complete
- **D.** BAC divided by CPI

124. Which project selection measure does NOT consider the time value of money?

- **A.** Simple payback period
- **B.** Discounted cash flow
- **C.** Net present value
- **D.** Internal rate of return

125. Project X has an IRR of 14% and Project Y has an IRR of 11%. The organisation's cost of capital is 12%. Which statement is correct?

- **A.** IRR cannot be compared with the cost of capital, so NPV must be used instead
- **B.** Both projects should be rejected, since neither IRR is at least 15%
- **C.** Project X exceeds the hurdle rate and is preferable; Project Y falls below it
- **D.** Project Y is preferable, since a lower IRR means a lower risk for the investor

126. A project's benefits are valued at $750,000 and its costs at $500,000 (both in present value). What is the benefit-cost ratio?

- **A.** 0.67
- **B.** 1.5
- **C.** 2.5
- **D.** 250,000

127. Which reserve is used when a previously identified risk occurs and its planned contingency response requires money?

- **A.** Management reserve
- **B.** Contingency reserve
- **C.** Sunk cost
- **D.** Opportunity cost

128. Which two actions are appropriate when the CPI drops to 0.82 midway through a project? (Choose two.)

- **A.** Reduce quality to save money without telling anyone
- **B.** Analyse the root causes of the cost overrun
- **C.** Develop corrective actions and present forecasts and options to the sponsor
- **D.** Hide the variance until the next phase
- **E.** Stop all work immediately

129. In an agile project, the team costs $60,000 per two-week sprint. The product owner estimates 9 more sprints are needed. What is the estimated cost to complete?

- **A.** $270,000
- **B.** $540,000
- **C.** $600,000
- **D.** $1,080,000

130. Which statement about funding requirements is correct?

- **A.** Funding is normally released in a single lump sum when the project charter is approved
- **B.** Funding requirements exclude management reserve, which is held outside the project budget
- **C.** Funds are often released in steps while spending is continuous, so they must be reconciled
- **D.** Funding requirements are derived directly from the activity list, not from the baseline

131. The cumulative cost baseline plotted over time typically forms what shape?

- **A.** Bell curve
- **B.** Pareto chart
- **C.** Straight vertical line
- **D.** S-curve

132. Which cost concept refers to costs that remain the same regardless of the volume of work?

- **A.** Fixed costs
- **B.** Variable costs
- **C.** Indirect costs
- **D.** Sunk costs

133. A project has BAC = $250,000, EV = $100,000 and AC = $125,000. What is the EAC assuming current cost performance continues? (In dollars.)

(Fill in the blank.)

Answer: _____

134. An analyst wants to show how much the project's actual spending exceeds the planned value of work completed so far. Which measure is this?

- **A.** VAC
- **B.** Cost variance
- **C.** Schedule variance
- **D.** TCPI

135. What is life-cycle costing?

- **A.** Recording costs that have already been incurred so that they can be recovered later
- **B.** Considering the total cost of ownership, from acquisition through operation to disposal
- **C.** Estimating the cost of each phase of the project life cycle separately from all of the others
- **D.** Calculating the cost of every sprint so that the burn rate can be tracked accurately

Answers — Part 6 (Finance)

Q116 — Answer: B

Correct: B. Level of accuracy

Level of accuracy defines the acceptable range of estimates; level of precision defines rounding (e.g., to the nearest $100).

Q117 — Answer: D

Correct: D. A share of the company's office rent and utilities

Indirect costs are shared overheads allocated to several projects. The other items are direct costs.

Q118 — Answer: D

Correct: D. Project budget = cost baseline + management reserve

The cost baseline includes contingency reserves; adding management reserve gives the project budget.

Q119 — Answer: B

Correct: B. 0.89

EV = 40% × 400,000 = 160,000. CPI = EV / AC = 160,000 / 180,000 ≈ 0.89.

Q120 — Answer: A

Correct: A. 0.80

PV = 50% × 400,000 = 200,000. SPI = EV / PV = 160,000 / 200,000 = 0.80.

Q121 — Answer: D

Correct: D. About $1,277,800

EAC = AC + (BAC − EV) / (CPI × SPI) = 444,444 + 600,000 / 0.72 = 444,444 + 833,333 ≈ $1,277,800.

Q122 — Answer: C

Correct: C. The project is expected to finish $50,000 over budget

VAC = BAC − EAC; a negative value means the forecast total cost exceeds the budget.

Q123 — Answer: C

Correct: C. AC plus a new bottom-up estimate to complete

When the original basis is flawed, the remaining work is re-estimated from the bottom up.

Q124 — Answer: A

Correct: A. Simple payback period

Simple payback ignores the time value of money and cash flows after the payback point. NPV, IRR and DCF discount future cash flows.

Q125 — Answer: C

Correct: C. Project X exceeds the hurdle rate and is preferable; Project Y falls below it

A project whose IRR exceeds the cost of capital creates value; higher IRR is better.

Q126 — Answer: B

Correct: B. 1.5

BCR = benefits / costs = 750,000 / 500,000 = 1.5, which is favourable because it exceeds 1.

Q127 — Answer: B

Correct: B. Contingency reserve

Contingency reserves cover identified risks (known-unknowns). Management reserve covers unidentified risks.

Q128 — Answer: B, C

Correct: B. Analyse the root causes of the cost overrun / C. Develop corrective actions and present forecasts and options to the sponsor

Variance analysis and transparent forecasting with options are appropriate. Hiding variances or secretly cutting quality violates ethics and governance.

Q129 — Answer: B

Correct: B. $540,000

Burn-rate forecasting: 9 sprints × $60,000 = $540,000.

Q130 — Answer: C

Correct: C. Funds are often released in steps while spending is continuous, so they must be reconciled

Funds are often released in steps; funding limit reconciliation ensures expenditure stays within available funds.

Q131 — Answer: D

Correct: D. S-curve

Spending is slow at the start, accelerates during execution and slows near completion, creating an S-shaped curve.

Q132 — Answer: A

Correct: A. Fixed costs

Fixed costs do not change with volume, such as set-up costs. Variable costs change with the amount of work.

Q133 — Answer: $312,500

CPI = 100,000 / 125,000 = 0.8. EAC = BAC / CPI = 250,000 / 0.8 = $312,500.

Q134 — Answer: B

Correct: B. Cost variance

CV = EV − AC compares the value of completed work with its actual cost.

Q135 — Answer: B

Correct: B. Considering the total cost of ownership, from acquisition through operation to disposal

Life-cycle costing supports holistic and sustainable decisions by looking beyond the project's delivery cost.

Part 7 — Stakeholders and Communications

136. When should stakeholder identification begin?

- **A.** Only when a stakeholder raises a concern or requests a change to the project
- **B.** As early as possible, usually once the charter is approved, and repeatedly afterwards
- **C.** After the project management plan is approved, when all of the stakeholders' roles are clear
- **D.** During planning, once the requirements have been collected and documented

137. A stakeholder has low power but high interest — for example, a group of end users. What is the usual strategy?

- **A.** Ignore
- **B.** Manage closely
- **C.** Keep satisfied
- **D.** Keep informed

138. Which communication method is most appropriate for an announcement that must reach all 2,000 employees and confirm delivery?

- **A.** Informal communication, through managers' hallway conversations
- **B.** Pull communication, such as a notice on the intranet home page
- **C.** Interactive communication, through one meeting per department
- **D.** Push communication, such as an email sent to all employees

139. What does the communication model term "noise" refer to?

- **A.** The total number of communication channels in the project
- **B.** Anything that interferes with sending or understanding a message
- **C.** The feedback the receiver returns to confirm understanding
- **D.** The loudness of the speaker's voice during a presentation or a meeting

140. A project team has 15 members including the project manager. How many communication channels exist?

(Fill in the blank.)

Answer: _____

141. Which document records each stakeholder's interests, influence, expectations and classification?

- **A.** Stakeholder engagement plan
- **B.** Stakeholder register
- **C.** Issue log
- **D.** Communications management plan

142. A powerful department head is resistant to the project. The project manager's analysis shows she fears losing headcount. What is the best approach?

- **A.** Raise her resistance in the steering meeting so others can pressure her
- **B.** Meet her to understand her concerns and explore how the project can address them
- **C.** Limit her involvement until implementation, so that her resistance does not spread to others

- **D.** Ask the sponsor to instruct her to support the project and stop objecting

143. Which stakeholder analysis model considers power, urgency and legitimacy?

- **A.** Stakeholder cube
- **B.** Power/interest grid
- **C.** Engagement assessment matrix
- **D.** Salience model

144. A stakeholder engagement assessment matrix shows a stakeholder as "Unaware" (current) and "Supportive" (desired). What should be the first engagement objective?

- **A.** Escalate to the sponsor to obtain the stakeholder's support
- **B.** Keep the stakeholder unaware to avoid creating resistance
- **C.** Invite the stakeholder to lead the project's steering committee from now on
- **D.** Make the stakeholder aware of the project and its impact on them

145. The sponsor complains that the status reports contain too much detail and asks for a one-page summary. What should the project manager do?

- **A.** Continue sending the detailed reports, since they contain everything the sponsor might need
- **B.** Send the detailed reports more often so the sponsor gets used to the level of detail
- **C.** Agree on a summary dashboard and update the communications management plan accordingly
- **D.** Stop sending reports to the sponsor and rely on the monthly steering committee instead

146. Which two practices support communication in a distributed agile team? (Choose two.)

- **A.** Holding the daily scrum only for co-located members
- **B.** Agreed overlap hours for synchronous collaboration
- **C.** Removing video from all calls
- **D.** Replacing all conversations with long written reports
- **E.** Shared digital task boards visible to all members

147. Which communication dimension is used when a project manager discusses a sensitive personnel issue with a team member?

- **A.** Pull communication on a wiki
- **B.** Push email to the entire team
- **C.** Formal written report sent to all of the stakeholders
- **D.** Informal, private, interactive conversation

148. A stakeholder suddenly opposes a decision that was agreed in a meeting last month. What could best have prevented misunderstanding?

- **A.** Making the decision without the stakeholders, so that later disagreements are avoided
- **B.** Documenting decisions and actions and circulating minutes for confirmation
- **C.** Holding fewer meetings so that decisions are not revisited so often
- **D.** Communicating decisions only verbally to keep the process flexible

149. Which process verifies that communications are having the intended effect and meeting stakeholders' information needs?

- **A.** Identify Stakeholders

- **B.** Manage Communications
- **C.** Monitor Communications
- **D.** Plan Communications Management

150. The project manager learns that a key user group in another country prefers indirect communication and avoids open disagreement in meetings. What should the project manager do?

- **A.** Exclude them from requirements sessions and send them the final documents
- **B.** Insist that they voice disagreement openly in the large requirements meetings
- **C.** Adapt how input is gathered, for example individual or anonymous feedback
- **D.** Interpret their silence in meetings as agreement with the proposed requirements

151. Which of the following is an example of pull communication?

- **A.** A telephone call to update the sponsor on a key decision
- **B.** A video conference with the steering committee members
- **C.** A knowledge repository that stakeholders consult when they need it
- **D.** A weekly status email sent to everyone on the project's distribution list

152. Which stakeholder direction of influence describes relationships with peer project managers competing for the same resources?

- **A.** Sideward
- **B.** Outward
- **C.** Upward
- **D.** Downward

153. A regulator is an important stakeholder whose approval is required. Which engagement approach is most appropriate?

- **A.** Treat the regulator as a low priority, since regulators are external parties
- **B.** Ask the team to interpret the regulations themselves to save time and cost
- **C.** Engage early and regularly to understand requirements and avoid surprises
- **D.** Contact the regulator once the product is complete and ready for approval

154. What is the project manager's main goal when managing stakeholder expectations?

- **A.** Keep expectations aligned with what can realistically be delivered
- **B.** Agree to stakeholders' requests so that satisfaction remains high
- **C.** Leave expectation management to the sponsor, who owns the business case
- **D.** Minimise contact with stakeholders so that fewer new requests arise

155. Which technique helps a speaker confirm that a listener understood a complex message?

- **A.** Sending the same message twice by email, a day apart
- **B.** Avoiding questions
- **C.** Speaking faster
- **D.** Asking the listener to paraphrase the key points

Answers — Part 7 (Stakeholders and Communications)

Q136 — Answer: B

Correct: B. As early as possible, usually once the charter is approved, and repeatedly afterwards

Identify Stakeholders is an initiating process, and identification is repeated throughout as the project and environment change.

Q137 — Answer: D

Correct: D. Keep informed

Low power, high interest stakeholders are kept informed; they can also become important allies.

Q138 — Answer: D

Correct: D. Push communication, such as an email sent to all employees

Push communication is sent to specific recipients and ensures distribution, although not necessarily understanding.

Q139 — Answer: B

Correct: B. Anything that interferes with sending or understanding a message

Noise includes distance, technology problems, language, culture, distractions and other barriers.

Q140 — Answer: 105

Channels = n(n − 1) / 2 = 15 × 14 / 2 = 105.

Q141 — Answer: B

Correct: B. Stakeholder register

The register contains identification, assessment and classification information. The engagement plan describes strategies to involve stakeholders.

Q142 — Answer: B

Correct: B. Meet her to understand her concerns and explore how the project can address them

Understanding interests and addressing concerns is the most effective way to move a resistant stakeholder toward support.

Q143 — Answer: D

Correct: D. Salience model

The salience model classifies stakeholders on these three attributes.

Q144 — Answer: D

Correct: D. Make the stakeholder aware of the project and its impact on them

Moving a stakeholder from unaware requires communication to build awareness before support can develop.

Q145 — Answer: C

Correct: C. Agree on a summary dashboard and update the communications management plan accordingly

Communication must meet stakeholder needs. Feedback triggers an update to the communications approach and plan.

Q146 — Answer: B, E

Correct: B. Agreed overlap hours for synchronous collaboration / E. Shared digital task boards visible to all members

Digital information radiators and overlapping working hours help distributed teams collaborate. Excluding remote members or relying only on reports weakens communication.

Q147 — Answer: D

Correct: D. Informal, private, interactive conversation

Sensitive issues are best handled privately and interactively, allowing dialogue and preserving dignity.

Q148 — Answer: B

Correct: B. Documenting decisions and actions and circulating minutes for confirmation

Documented and confirmed decisions reduce misunderstandings and provide a reference when disputes arise.

Q149 — Answer: C

Correct: C. Monitor Communications

Monitor Communications checks the effectiveness of communications and triggers adjustments.

Q150 — Answer: C

Correct: C. Adapt how input is gathered, for example individual or anonymous feedback

Cultural awareness means adapting methods so stakeholders can contribute in ways comfortable for them.

Q151 — Answer: C

Correct: C. A knowledge repository that stakeholders consult when they need it

Pull communication lets recipients access information at their discretion, suitable for large volumes or audiences.

Q152 — Answer: A

Correct: A. Sideward

Sideward influence concerns peers, such as other project managers or middle managers.

Q153 — Answer: C

Correct: C. Engage early and regularly to understand requirements and avoid surprises

Early engagement with high-power stakeholders such as regulators prevents costly late rework and delays.

Q154 — Answer: A

Correct: A. Keep expectations aligned with what can realistically be delivered

Expectation management is continuous: surface expectations, negotiate trade-offs and communicate transparently.

Q155 — Answer: D

Correct: D. Asking the listener to paraphrase the key points

Paraphrasing and feedback are part of active listening and confirm shared understanding.

Part 8 — Resources and Procurement

156. Which document shows a hierarchical list of resources by category and type, such as labour, equipment and materials?

- **A.** Risk breakdown structure
- **B.** Work breakdown structure
- **C.** Organisational breakdown structure
- **D.** Resource breakdown structure

157. A project needs a database expert who is fully booked by her functional manager for another initiative. What should the project manager do first?

- **A.** Negotiate with the functional manager to explore options such as partial allocation
- **B.** Remove the database work from scope and inform the sponsor of the change
- **C.** Ask the expert to support the project in her spare time without telling her manager
- **D.** Escalate to the CEO so the project receives priority over the other initiative

158. A team charter states that decisions are made by consensus and conflicts are first discussed between the people involved. A conflict arises. What should the project manager do?

- **A.** Rewrite the team charter so that the project manager resolves all conflicts from now on
- **B.** Decide the issue for the team immediately so that work can continue without delay
- **C.** Remind the team of the agreed conflict process in the charter and support them in using it
- **D.** Ignore the charter, since conflict resolution is entirely the functional managers' responsibility

159. Which method helps select team members by weighting criteria such as availability, cost, experience, skills and attitude?

- **A.** Multicriteria decision analysis
- **B.** Monte Carlo simulation
- **C.** Bidder conference
- **D.** Resource levelling

160. Which statement about virtual teams is correct?

- **A.** They remove the need for a communications plan, since everything happens online
- **B.** They widen the talent pool but need careful communication planning and tools
- **C.** They consistently perform worse than co-located teams on complex work
- **D.** They cannot use agile practices, which require all members in one room

161. During proposal evaluation, the buyer eliminates proposals that do not meet mandatory requirements before scoring the remainder. Which technique is this?

- **A.** Independent estimate
- **B.** Weighting
- **C.** Negotiation
- **D.** Screening

162. A seller's proposal price is 40% below the buyer's independent cost estimate. What should the buyer consider?

- **A.** Awarding the contract immediately, since the low price frees up budget for other important work
- **B.** Increasing the budget, since the independent estimate was clearly too high
- **C.** Rejecting the proposal at once, since low prices always signal poor quality
- **D.** Whether the seller misunderstood the statement of work before awarding the contract

163. Which contract clause excuses a party from performing obligations due to extraordinary events beyond its control, such as natural disasters?

- **A.** Force majeure
- **B.** Liquidated damages
- **C.** Warranty
- **D.** Retainage

164. A seller delivers late, and the contract contains a liquidated damages clause. What does this clause provide?

- **A.** A bonus paid to the seller for delivering ahead of the contract schedule
- **B.** Reimbursement of all of the legal fees that the buyer incurs in enforcing the contract
- **C.** Automatic termination of the contract when the delivery date is missed
- **D.** Compensation, agreed in advance, that the seller pays for the specified breach

165. The buyer terminates a contract because changed business priorities mean the work is no longer needed, as allowed by the contract. This is termination for:

- **A.** Convenience
- **B.** Force majeure
- **C.** Cause
- **D.** Default

166. CPIF contract: target cost $500,000, target fee $40,000, buyer/seller share ratio 70/30, actual cost $540,000. What is the total price the buyer pays? (In dollars.)

(Fill in the blank.)

Answer: _____

167. Which process monitors whether physical resources such as equipment and materials are available as planned and used efficiently?

- **A.** Lead the Team
- **B.** Monitor and Control Resourcing
- **C.** Estimate Resources
- **D.** Plan Resource Management

168. During a project, the team stays in the storming stage longer than expected, with frequent disagreements on approach. What should the project manager do?

- **A.** Wait for the team to reach norming on its own, since storming always ends naturally
- **B.** Forbid disagreements in meetings so the team can focus on delivering the work
- **C.** Replace the team members who disagree most often with more cooperative people

- **D.** Coach the team, facilitate constructive conflict resolution and revisit agreements

169. Which two practices are part of contract administration during execution? (Choose two.)

- **A.** Holding the bidder conference
- **B.** Writing the initial make-or-buy analysis
- **C.** Reviewing seller performance against the contract
- **D.** Approving payments after verifying completed work
- **E.** Selecting the contract type

170. A team member receives a public award that only one person can win each month. The project manager notices rising competition and less collaboration. What should be considered?

- **A.** Competition between team members always improves collaboration and output
- **B.** Recognition should be reserved for managers to avoid tension within the team
- **C.** All rewards should be removed, since recognition always creates competition
- **D.** Win-lose rewards can harm teamwork; team-based recognition may work better

171. Which statement about training team members is correct?

- **A.** Only the sponsor and project manager should receive training paid by the project
- **B.** Training can be a project cost when needed for the project, planned in the resource plan
- **C.** Training should only be delivered after closure, so it does not slow the project down
- **D.** Training is never a project cost, since it benefits the organisation long after the project

172. A seller claims additional payment for work it says the buyer requested verbally; the buyer disagrees. What is this called?

- **A.** A procurement audit
- **B.** A warranty
- **C.** A claim (dispute)
- **D.** A change log

173. What is the main purpose of a procurement audit?

- **A.** To select the seller whose proposal offers the best overall value to the buyer
- **B.** To write the procurement statement of work for the next contract
- **C.** To negotiate the final price and terms with the preferred seller
- **D.** To review the procurement process and capture lessons for future procurements

174. In an adaptive team, who decides how the work in the sprint backlog is distributed among the developers?

- **A.** The project manager, based on skills
- **B.** The product owner, based on priority
- **C.** The developers, through self-organisation
- **D.** The functional managers of each developer

175. Which RACI assignment is a problem?

- **A.** Three people marked Informed for an activity
- **B.** One person marked Responsible and Accountable for the same activity
- **C.** Two people marked Accountable for the same activity

- **D.** Two people marked Consulted for an activity

Answers — Part 8 (Resources and Procurement)

Q156 — Answer: D

Correct: D. Resource breakdown structure

The RBS organises resources by category and type. The OBS shows organisational units and the WBS shows deliverables.

Q157 — Answer: A

Correct: A. Negotiate with the functional manager to explore options such as partial allocation

Negotiation with the resource owner is the first step in a matrix environment. Escalation follows only if negotiation fails.

Q158 — Answer: C

Correct: C. Remind the team of the agreed conflict process in the charter and support them in using it

The team charter is the team's own agreement; the PM reinforces it and facilitates as needed.

Q159 — Answer: A

Correct: A. Multicriteria decision analysis

Multicriteria decision analysis uses weighted criteria to compare candidates or options.

Q160 — Answer: B

Correct: B. They widen the talent pool but need careful communication planning and tools

Virtual teams offer access to skills and flexibility, but need clear communication protocols, tools and attention to time zones and culture.

Q161 — Answer: D

Correct: D. Screening

Screening filters out proposals that fail must-meet criteria; weighting then scores the remaining proposals.

Q162 — Answer: D

Correct: D. Whether the seller misunderstood the statement of work before awarding the contract

A large difference from the independent estimate may indicate that the SOW was unclear or the seller misunderstood it, creating risk.

Q163 — Answer: A

Correct: A. Force majeure

Force majeure covers unforeseeable, uncontrollable events. Liquidated damages are pre-agreed compensation for breaches such as late delivery.

Q164 — Answer: D

Correct: D. Compensation, agreed in advance, that the seller pays for the specified breach

Liquidated damages are an amount agreed in advance as compensation for specific breaches, avoiding the need to prove actual damages.

Q165 — Answer: A

Correct: A. Convenience

Termination for convenience is allowed when the contract permits the buyer to end it without a breach, usually with compensation for work done.

Q166 — Answer: $568,000

Overrun = $40,000; seller's share = 30% × 40,000 = $12,000; fee = 40,000 − 12,000 = $28,000; total = 540,000 + 28,000 = $568,000.

Q167 — Answer: B

Correct: B. Monitor and Control Resourcing

Monitor and Control Resourcing focuses on physical, virtual and externally sourced resources.

Q168 — Answer: D

Correct: D. Coach the team, facilitate constructive conflict resolution and revisit agreements

In storming, the leader coaches and facilitates; constructive conflict helps the team progress to norming.

Q169 — Answer: C, D

Correct: C. Reviewing seller performance against the contract / D. Approving payments after verifying completed work

Contract administration covers performance reviews, inspections, payments, changes and claims. Make-or-buy, contract type and bidder conferences belong to earlier procurement work.

Q170 — Answer: D

Correct: D. Win-lose rewards can harm teamwork; team-based recognition may work better

Zero-sum rewards can undermine collaboration; recognition should reinforce the desired team behaviours.

Q171 — Answer: B

Correct: B. Training can be a project cost when needed for the project, planned in the resource plan

Training is a team development tool and its costs can be part of the project when it supports the project's needs.

Q172 — Answer: C

Correct: C. A claim (dispute)

Claims are contested changes or disputes about compensation; they are administered through the contract's claims process.

Q173 — Answer: D

Correct: D. To review the procurement process and capture lessons for future procurements

Procurement audits examine the procurement process from planning to closure to improve future procurements.

Q174 — Answer: C

Correct: C. The developers, through self-organisation

Developers self-organise to decide who does what and how; the product owner decides what is most valuable.

Q175 — Answer: C

Correct: C. Two people marked Accountable for the same activity

Only one person should be Accountable for each activity. A person can be both Responsible and Accountable.

Part 9 — Risk

176. Which element of the risk management plan defines the scales used to rate probability and impact?

- **A.** Stakeholder risk appetite statements
- **B.** Risk register entries for each identified risk
- **C.** Definitions of risk probability and impact
- **D.** Risk breakdown structure categories

177. Which statement is written in the recommended cause–risk–effect format?

- **A.** Integration testing may be delayed by two weeks, which is a serious risk that the vendor needs to manage carefully
- **B.** Because the vendor is new to us, delivery of components may be late, which would delay integration testing by two weeks
- **C.** The vendor is new to us and has not yet delivered any components, so integration testing is a major concern for the team
- **D.** Late delivery of components by the vendor is an issue that will cause problems in testing and must be escalated

178. A risk is rated with high probability and high impact but its effect will not be felt for eight months. Which risk parameter captures how soon the risk could affect objectives?

- **A.** Proximity
- **B.** Manageability
- **C.** Dormancy
- **D.** Detectability

179. Which technique helps reduce bias by gathering expert opinions anonymously over several rounds until they converge?

- **A.** SWOT analysis
- **B.** Delphi technique
- **C.** Brainstorming
- **D.** Nominal group technique

180. Which output summarises the sources of overall project risk and the status of individual risks for stakeholders?

- **A.** Risk report
- **B.** Risk breakdown structure
- **C.** Risk register
- **D.** Assumption log

181. To reduce the probability that a new technology will fail in production, the team builds a prototype and runs a pilot. Which threat response is this?

- **A.** Accept
- **B.** Avoid
- **C.** Transfer

- **D.** Mitigate

182. The team decides to use a proven technology instead of a new one to eliminate a technical threat. Which response is this?

- **A.** Escalate
- **B.** Mitigate
- **C.** Exploit
- **D.** Avoid

183. Two companies form a joint venture to win a contract that neither could win alone. Which opportunity response is this?

- **A.** Enhance
- **B.** Share
- **C.** Transfer
- **D.** Exploit

184. The project sets aside two weeks of schedule reserve to cope with possible weather delays and takes no other action. Which strategy is this?

- **A.** Active acceptance
- **B.** Mitigation
- **C.** Avoidance
- **D.** Passive acceptance

185. After a vendor contract with penalty clauses was signed to transfer schedule risk, the vendor raised its prices for the next phase. How is this new risk of higher costs classified?

- **A.** Residual risk
- **B.** Secondary risk
- **C.** Trigger
- **D.** Workaround

186. Which quantitative technique runs thousands of iterations using probability distributions to estimate the likelihood of meeting a target date?

- **A.** Monte Carlo simulation
- **B.** Probability and impact matrix
- **C.** Checklist analysis
- **D.** SWOT analysis

187. A decision tree shows Option A with an expected value of +40,
000*andOptionBwithanexpectedvalueof*+55,000. Both are opportunities. Which option should be chosen based on EMV?

- **A.** Option B
- **B.** Option A
- **C.** Neither, because decision trees are qualitative
- **D.** Both, combined

188. Who is responsible for implementing the agreed response when a risk's trigger occurs?

- **A.** The sponsor
- **B.** The procurement department
- **C.** The customer
- **D.** The risk owner

189. During a risk review, the team finds several risks that are no longer relevant. What should happen?

- **A.** Close them and consider releasing the related contingency reserve as the plan allows
- **B.** Keep them open for the rest of the project in case the conditions change again
- **C.** Transfer them to the customer, who is better placed to watch for the conditions that trigger them
- **D.** Delete them from the register so that the document stays short and readable

190. Which two techniques are typically used in qualitative risk analysis? (Choose two.)

- **A.** Decision tree with EMV
- **B.** Monte Carlo simulation
- **C.** Assessment of risk urgency and proximity
- **D.** Probability and impact matrix
- **E.** Tornado diagram from sensitivity analysis

191. A project manager identifies a risk that a sister program might reprioritise shared resources. The project cannot influence this decision. What is the best response?

- **A.** Escalate it to the program or portfolio manager who owns the decision
- **B.** Accept it passively, since the project cannot influence the decision
- **C.** Transfer it by buying insurance against the loss of shared resources
- **D.** Mitigate it by hiring new staff now in case the resources are withdrawn

192. In an adaptive project, how are high-risk backlog items usually handled?

- **A.** They are handed to the sponsor, who owns all product risks
- **B.** They are often prioritised early to reduce uncertainty quickly
- **C.** They are removed from the backlog until the risk has disappeared
- **D.** They are postponed to the final sprint when the team is most experienced

193. Which document lists assumptions whose possible inaccuracy should be examined as a source of risk?

- **A.** Change log
- **B.** Team charter
- **C.** Assumption log
- **D.** Stakeholder register

194. What is the main difference between a contingency plan and a fallback plan?

- **A.** A contingency plan handles issues, while a fallback plan handles opportunities
- **B.** Both terms describe the same plan, prepared for risks above the threshold
- **C.** A contingency plan runs when a trigger occurs; a fallback plan is used if it fails
- **D.** A fallback plan runs when a trigger occurs; a contingency plan is used if it fails

195. A project has an aggressive risk appetite, while its client is very risk-averse. What should the project manager do?

- **A.** Surface the differences with key stakeholders and agree thresholds in the risk management plan
- **B.** Remove the client from risk discussions so that the team can work without pressure
- **C.** Ignore risk appetite, since it is subjective and cannot be measured or documented
- **D.** Follow the project team's appetite, since the team is responsible for delivering all of the work

Answers — Part 9 (Risk)

Q176 — Answer: C

Correct: C. Definitions of risk probability and impact

The risk management plan contains probability and impact definitions and the probability-impact matrix used in analysis.

Q177 — Answer: B

Correct: B. Because the vendor is new to us, delivery of components may be late, which would delay integration testing by two weeks

A good risk statement links a cause, an uncertain event and its effect on objectives.

Q178 — Answer: A

Correct: A. Proximity

Proximity is the time before the risk could affect objectives. Dormancy is the time between occurrence and discovery of its effect.

Q179 — Answer: B

Correct: B. Delphi technique

The Delphi technique uses anonymous, iterative rounds with a facilitator to reach consensus without dominance by individuals.

Q180 — Answer: A

Correct: A. Risk report

The risk report presents overall project risk and summary information on individual risks; the register lists individual risks in detail.

Q181 — Answer: D

Correct: D. Mitigate

Mitigation reduces probability and/or impact. Avoiding would remove the threat entirely, for example by not using the technology.

Q182 — Answer: D

Correct: D. Avoid

Changing the plan to eliminate the threat is avoidance.

Q183 — Answer: B

Correct: B. Share

Sharing allocates ownership of an opportunity to a party better able to capture it, such as a partner.

Q184 — Answer: A

Correct: A. Active acceptance

Active acceptance establishes a contingency reserve. Passive acceptance takes no action except periodic review.

Q185 — Answer: B

Correct: B. Secondary risk

A secondary risk arises directly from implementing a risk response.

Q186 — Answer: A

Correct: A. Monte Carlo simulation

Monte Carlo simulation models combined uncertainty to produce probability distributions of outcomes.

Q187 — Answer: A

Correct: A. Option B

For opportunities, choose the higher expected value; for costs, choose the lower expected cost.

Q188 — Answer: D

Correct: D. The risk owner

Each risk has a risk owner responsible for monitoring it and implementing the response.

Q189 — Answer: A

Correct: A. Close them and consider releasing the related contingency reserve as the plan allows

Risk reassessment closes outdated risks; reserve analysis may release unneeded reserves.

Q190 — Answer: C, D

Correct: C. Assessment of risk urgency and proximity / D. Probability and impact matrix

Qualitative analysis prioritises risks using probability, impact and other parameters. Simulation, EMV decision trees and sensitivity analysis are quantitative.

Q191 — Answer: A

Correct: A. Escalate it to the program or portfolio manager who owns the decision

The risk lies outside the project's authority, so it should be escalated to the appropriate owner.

Q192 — Answer: B

Correct: B. They are often prioritised early to reduce uncertainty quickly

Tackling risky items early validates assumptions and reduces risk exposure; risk responses can be added to the backlog.

Q193 — Answer: C

Correct: C. Assumption log

Assumption and constraint analysis examines the assumption log for assumptions that might be inaccurate, unstable or incomplete.

Q194 — Answer: C

Correct: C. A contingency plan runs when a trigger occurs; a fallback plan is used if it fails

Fallback plans are backups for when the primary contingent response is insufficient.

Q195 — Answer: A

Correct: A. Surface the differences with key stakeholders and agree thresholds in the risk management plan

Stakeholder risk appetites shape thresholds; differences should be surfaced and agreed early.

Part 10 — Agile and Hybrid Delivery

196. Which Agile Manifesto principle is reflected when a team delivers working features every two weeks?

- **A.** Business people and developers must work together daily throughout the project
- **B.** Deliver working software frequently, preferring the shorter timescale
- **C.** Build projects around motivated individuals and trust them to get the job done
- **D.** Agile processes promote sustainable development at a constant pace indefinitely

197. What is the primary measure of progress according to the Agile Manifesto principles?

- **A.** Hours worked by the team
- **B.** Percentage of documentation completed
- **C.** Number of meetings held
- **D.** Working software (a working product)

198. The developers want to change their workflow for the next sprint. Which event is the most appropriate place to agree on the change?

- **A.** Backlog refinement
- **B.** Sprint review
- **C.** Daily scrum
- **D.** Sprint retrospective

199. During sprint planning, who decides how many backlog items the developers can complete in the sprint?

- **A.** The project sponsor
- **B.** The developers
- **C.** The product owner
- **D.** The scrum master

200. A product owner is rarely available, and the team is waiting for decisions. What should the scrum master do first?

- **A.** Cancel the sprint and restart once the product owner has more time available
- **B.** Ask the developers to assume the requirements and correct them in a later sprint
- **C.** Make the product decisions on the product owner's behalf to keep the team moving
- **D.** Discuss the impact with the product owner and agree how to ensure availability

201. Which artifact contains the sprint goal, the selected items and the plan to deliver them?

- **A.** Product backlog
- **B.** Sprint backlog
- **C.** Increment
- **D.** Release plan

202. What is the commitment associated with the increment in Scrum?

- **A.** Definition of done

- **B.** Velocity target
- **C.** Product goal
- **D.** Sprint goal

203. A Kanban board shows 9 items in the "Review" column, which has a WIP limit of 4. What should the team do?

- **A.** Stop starting new work and swarm to clear the review bottleneck
- **B.** Start more development work so that no developer is left idle
- **C.** Remove the review column and let developers review their own work
- **D.** Raise the review column's WIP limit to 10 to reflect the current reality

204. Which XP practice ensures that code from all developers is merged and tested frequently, often several times a day?

- **A.** Continuous integration
- **B.** Planning game
- **C.** Sustainable pace
- **D.** Collective code ownership

205. In SAFe, what is the name of the regular event where all teams on an agile release train plan together?

- **A.** Sprint review
- **B.** Bidder conference
- **C.** PI (program increment) planning
- **D.** Scrum of scrums

206. A team's velocity has been 30, 32 and 31. A manager asks the team to commit to 45 points next sprint. What should the scrum master do?

- **A.** Agree to the target and ask the team to work overtime until all of the extra points are delivered
- **B.** Re-estimate the stories with larger point values so that the team reaches 45 points
- **C.** Explain that velocity reflects sustainable capacity, and involve the product owner in scope
- **D.** Relax the definition of done for one sprint so the team can complete more stories

207. What is the main purpose of the sprint review?

- **A.** Approve the budget for the next sprint with the project sponsor
- **B.** Inspect the increment with stakeholders and adapt the product backlog
- **C.** Agree on improvements to the team's way of working and tools
- **D.** Evaluate each developer's individual contribution to the work done during the sprint

208. Which two are values of Scrum? (Choose two.)

- **A.** Openness
- **B.** Documentation
- **C.** Competition
- **D.** Courage
- **E.** Hierarchy

209. A team using Scrum notices that stories often remain "almost done" at the end of the sprint. Which action best addresses this?

- **A.** Remove testing from the definition of done so that stories can be closed faster
- **B.** Extend the sprint by a few days whenever stories are close to completion
- **C.** Count the "almost done" stories toward velocity to reflect the team's real effort
- **D.** Limit WIP and slice stories smaller so they can be finished within the sprint

210. Which agile contract option allows the customer to stop the project when enough value has been delivered, typically by paying a predefined fee?

- **A.** Firm fixed price for the entire scope
- **B.** Graduated fixed price
- **C.** Early cancellation option
- **D.** Not-to-exceed time and materials

211. In a hybrid project, the software team uses sprints while procurement of hardware follows a fixed contract with delivery milestones. What is the project manager's key integration responsibility?

- **A.** Let both parts run independently, and integrate them only at the very end of the project
- **B.** Ask the hardware vendor to deliver in two-week sprints to match the software team
- **C.** Ask the software team to stop using sprints so that both parts follow one schedule
- **D.** Align release plans with hardware milestones and manage the dependencies between them

212. What is the role of the project manager in an agile team according to modern practice?

- **A.** Servant leader who facilitates, removes impediments and coordinates with the organisation
- **B.** Technical lead who decides the architecture and design for the development team
- **C.** Decision-maker who orders the product backlog on behalf of the stakeholders
- **D.** Coordinator who assigns the daily tasks to developers to keep the sprint plan on track at all times

213. Which Lean waste is reduced by keeping the same team together across projects?

- **A.** Extra processing of invoices
- **B.** Overproduction only
- **C.** Relearning and handoffs
- **D.** Transportation of materials

214. What does a definition of ready help the team achieve?

- **A.** Decide when an increment is ready to be released to production users
- **B.** Measure customer satisfaction with each feature after it has been released to the users
- **C.** Approve change requests before they are added to the product backlog
- **D.** Make sure backlog items are clear and small enough before they enter a sprint

215. A stakeholder asks why the agile team won't commit to the full feature list for a fixed release date. What is the best explanation?

- **A.** Agile teams avoid commitments of any kind, so no feature list can ever be promised
- **B.** With a fixed date and team, scope is flexible, so the most valuable features come first
- **C.** The team prefers not to work under pressure, so it keeps its commitments small
- **D.** The release date will be moved automatically until all of the listed features have been completed

216. Which approach coordinates multiple Scrum teams through a regular meeting of representatives who discuss dependencies and impediments?

- **A.** Sprint retrospective
- **B.** Daily scrum of one team
- **C.** Product box exercise
- **D.** Scrum of scrums

217. In Kanban, which statement about roles and iterations is correct?

- **A.** Kanban requires a scrum master and fixed two-week iterations to manage the flow of work
- **B.** Kanban prescribes no roles or fixed iterations and starts from the current process
- **C.** Kanban forbids work-in-progress limits, because they reduce team utilisation
- **D.** Kanban requires a product owner to accept each card before it can be moved

218. Match each agile concept with its description.

(Matching: pair every numbered item with one lettered option.)

Item	Option
1. Spike	**A.** Timeboxed experiment to reduce uncertainty
2. MVP	**B.** Amount of work completed per iteration
3. Information radiator	**C.** Smallest product used to validate a key assumption with users
4. Velocity	**D.** Highly visible display of project information

219. Which agile practice best embodies "simplicity — maximising the amount of work not done"?

- **A.** Building only what the current highest-priority needs require
- **B.** Designing for all imaginable future scenarios to keep options open
- **C.** Writing complete documentation up front so nothing has to be reworked
- **D.** Including every possible feature in the first release to avoid later changes

Answers — Part 10 (Agile and Hybrid Delivery)

Q196 — Answer: B

Correct: B. Deliver working software frequently, preferring the shorter timescale

Frequent delivery of working software in short timescales is a core manifesto principle.

Q197 — Answer: D

Correct: D. Working software (a working product)

Working software is the primary measure of progress.

Q198 — Answer: D

Correct: D. Sprint retrospective

The retrospective inspects the team's process and plans improvements.

Q199 — Answer: B

Correct: B. The developers

Developers assess their capacity and decide how much they can take on. The product owner clarifies priorities and value.

Q200 — Answer: D

Correct: D. Discuss the impact with the product owner and agree how to ensure availability

The scrum master coaches the PO and works to resolve the impediment; taking over the PO's decisions is inappropriate.

Q201 — Answer: B

Correct: B. Sprint backlog

The sprint backlog includes the sprint goal (its commitment), the selected product backlog items and the plan.

Q202 — Answer: A

Correct: A. Definition of done

Each Scrum artifact has a commitment: product backlog—product goal; sprint backlog—sprint goal; increment—definition of done.

Q203 — Answer: A

Correct: A. Stop starting new work and swarm to clear the review bottleneck

Respecting WIP limits means finishing work before starting more; the team collaborates to clear the bottleneck.

Q204 — Answer: A

Correct: A. Continuous integration

Continuous integration merges and tests code frequently to detect integration problems early.

Q205 — Answer: C

Correct: C. PI (program increment) planning

PI planning is a face-to-face (or virtual) event where teams on an ART align on objectives for the next increment.

Q206 — Answer: C

Correct: C. Explain that velocity reflects sustainable capacity, and involve the product owner in scope

Velocity is a planning tool, not a target. Forcing higher commitments damages quality, sustainability and trust.

Q207 — Answer: B

Correct: B. Inspect the increment with stakeholders and adapt the product backlog

The sprint review gathers feedback on the product and adjusts the backlog; the retrospective handles process improvements.

Q208 — Answer: A, D

Correct: A. Openness / D. Courage

The Scrum values are commitment, focus, openness, respect and courage.

Q209 — Answer: D

Correct: D. Limit WIP and slice stories smaller so they can be finished within the sprint

Smaller stories and lower WIP improve flow and completion. Extending sprints or counting unfinished work hides problems.

Q210 — Answer: C

Correct: C. Early cancellation option

The early cancellation option lets the customer end the contract early, usually with a cancellation fee, when sufficient value has been delivered.

Q211 — Answer: D

Correct: D. Align release plans with hardware milestones and manage the dependencies between them

Hybrid integration aligns plans at integration points and manages cross-component dependencies and risks.

Q212 — Answer: A

Correct: A. Servant leader who facilitates, removes impediments and coordinates with the organisation

Agile project managers serve and enable the team; product decisions belong to the product owner and technical decisions to the team.

Q213 — Answer: C

Correct: C. Relearning and handoffs

Stable teams retain knowledge, reducing relearning and the loss that occurs during handoffs.

Q214 — Answer: D

Correct: D. Make sure backlog items are clear and small enough before they enter a sprint

A DoR sets criteria (clear acceptance criteria, small size, dependencies resolved) for items to be pulled into a sprint.

Q215 — Answer: B

Correct: B. With a fixed date and team, scope is flexible, so the most valuable features come first

Agile planning often fixes time and resources and varies scope, delivering the highest value first.

Q216 — Answer: D

Correct: D. Scrum of scrums

A scrum of scrums brings team representatives together to coordinate integration and remove cross-team impediments.

Q217 — Answer: B

Correct: B. Kanban prescribes no roles or fixed iterations and starts from the current process

Kanban is evolutionary: visualise work, limit WIP, manage flow, make policies explicit and improve collaboratively.

Q218 — Answer: 1→A, 2→C, 3→D, 4→B

Correct pairs: 1. Spike → Timeboxed experiment to reduce uncertainty; 2. MVP → Smallest product used to validate a key assumption with users; 3. Information radiator → Highly visible display of project information; 4. Velocity → Amount of work completed per iteration

Spikes reduce uncertainty, MVPs validate value hypotheses, information radiators make information visible and velocity measures completed work per iteration.

Q219 — Answer: A

Correct: A. Building only what the current highest-priority needs require

Simplicity avoids unnecessary work, such as speculative features and premature design.

Part 11 — Leading People and Ethics

220. A newly hired engineer is enthusiastic but has little experience with the company's tools. According to situational leadership, which style fits best?

- **A.** Delegating — low direction and low support
- **B.** Supporting — low direction and high support
- **C.** Directing — high direction and low support
- **D.** Laissez-faire — no direction

221. Which leadership style focuses on inspiring followers through a compelling vision, intellectual stimulation and individual consideration?

- **A.** Autocratic
- **B.** Transformational
- **C.** Laissez-faire
- **D.** Transactional

222. A team member respects the project manager because of her deep technical knowledge. Which power is she using?

- **A.** Expert power
- **B.** Referent power
- **C.** Legitimate power
- **D.** Reward power

223. According to McGregor, a manager who believes employees are self-motivated and seek responsibility holds which view?

- **A.** Expectancy theory
- **B.** Theory Z
- **C.** Theory Y
- **D.** Theory X

224. According to Maslow, a team member worried about losing her job is focused on which level of needs?

- **A.** Self-actualisation
- **B.** Esteem
- **C.** Belonging
- **D.** Safety

225. Two senior engineers have a long-running conflict that affects the team. Private discussions facilitated by the project manager have not worked. What is the most appropriate next step?

- **A.** Ignore it, since some conflict between senior people is normal and healthy
- **B.** Involve their functional managers or HR according to organisational policy
- **C.** Remove both engineers from the project immediately without further discussion
- **D.** Reprimand both engineers publicly so that the team sees the issue is serious

226. Which conflict resolution technique is described as finding solutions that bring some degree of satisfaction to all parties by each giving up something?

- **A.** Force/direct
- **B.** Smooth/accommodate
- **C.** Collaborate/problem-solve
- **D.** Compromise/reconcile

227. Which is the most common source of conflict on projects according to classic research?

- **A.** Company logos
- **B.** Personalities
- **C.** Office space
- **D.** Schedules

228. A project manager stays calm and thinks before responding when a stakeholder angrily criticises the team. Which emotional intelligence component is she demonstrating?

- **A.** Self-regulation
- **B.** Motivation
- **C.** Social skills
- **D.** Empathy

229. A team uses "fist of five" to check agreement on a decision. Two members show two fingers. What should happen?

- **A.** Discuss their concerns before finalising the decision
- **B.** Proceed, since a clear majority supports the decision
- **C.** Ask the project manager to make the final decision alone
- **D.** Exclude the two members from this particular decision

230. In negotiations with a vendor, the project manager has identified that another supplier can deliver the same service for $90,000. What is this alternative called?

- **A.** Point of total assumption
- **B.** ZOPA
- **C.** BATNA
- **D.** Sunk cost

231. A vendor offers the project manager free tickets to a major sporting event during an open tender. What should the project manager do?

- **A.** Accept the tickets on behalf of the team as a reward for their hard work
- **B.** Accept, and make sure the evaluation remains objective despite the gift
- **C.** Accept quietly, since such invitations are a normal business courtesy
- **D.** Decline the offer, follow company policy and disclose it as appropriate

232. The project manager discovers that a team member reported tests as passed when they were not run. What should the project manager do first?

- **A.** Address it with the team member, correct the records and follow the organisation's procedures
- **B.** Run the tests quietly and keep the matter private to protect the team member's reputation

- **C.** Report it straight to the customer before discussing it with anyone inside the organisation first
- **D.** Ignore it for now, since the deadline is close and the tests can be run after release

233. Which statement about the PMI Code of Ethics is correct?

- **A.** It is a set of optional recommendations that certification holders may choose to follow if they wish
- **B.** It addresses financial matters such as fees and budgets, not professional behaviour
- **C.** It sets aspirational and mandatory standards for responsibility, respect, fairness and honesty
- **D.** It applies only to PMI employees and volunteers, not to certification holders

234. A team member is consistently late on deliverables. In a private conversation, she reveals she is caring for a sick parent. What should the project manager do?

- **A.** Report her to the functional manager for poor performance so it is documented
- **B.** Ignore the situation, since personal matters should not affect any of the project's decisions
- **C.** Reassign all her work to others immediately without discussing it with her
- **D.** Show empathy and explore flexible arrangements while agreeing realistic commitments

235. In the Tuckman model, which stage is characterised by the team becoming interdependent and working through issues smoothly with little supervision?

- **A.** Forming
- **B.** Performing
- **C.** Storming
- **D.** Norming

236. Which motivator, according to Daniel Pink, refers to the desire to get better at something that matters?

- **A.** Hygiene
- **B.** Autonomy
- **C.** Purpose
- **D.** Mastery

237. Which two actions build psychological safety in a team? (Choose two.)

- **A.** Publicly naming people responsible for defects
- **B.** Treating mistakes as learning opportunities in retrospectives
- **C.** Making all decisions without discussion
- **D.** Punishing team members who disagree with the plan
- **E.** Thanking people who raise problems early

238. A project manager notices that a quiet team member's ideas are often ignored in meetings dominated by two senior members. What is the best action?

- **A.** Ask the quiet member to send ideas by email instead of attending the meetings
- **B.** Make the decisions alone so that no one dominates the team's discussions
- **C.** Let the senior members lead the discussions, since their experience is the most valuable to the team
- **D.** Facilitate meetings so everyone contributes, for example by round-robin or written input

239. Which statement best describes the difference between leadership and management?

- **A.** Management focuses on processes and control; leadership on vision and people
- **B.** Management is unnecessary in agile projects, where only leadership is needed
- **C.** Leadership is reserved for executives; management is the role of project managers
- **D.** Leadership and management are two names for the same set of activities

240. A project manager working in a country where facilitation payments are customary is asked to pay an official to speed up a permit. Company policy and international anti-bribery laws prohibit it. What should the project manager do?

- **A.** Refuse, follow the law and company policy, and seek legitimate ways to speed up the permit
- **B.** Ask a local subcontractor to make the payment and include it in their invoice
- **C.** Pay it, since it is a local custom and the amount is small compared with the cost of the delay
- **D.** Pay it personally, so that the company itself is not involved in the transaction

241. Which coaching model uses the steps Outcome, Situation, Choices and consequences, Actions, and Review?

- **A.** DMAIC
- **B.** ADKAR
- **C.** SWOT
- **D.** OSCAR

242. The team successfully delivered a difficult release. What is the most appropriate recognition approach?

- **A.** A general thank-you message at the end of the project, months later
- **B.** No recognition, since delivering releases is part of everyone's job
- **C.** A reward for the project manager, who is accountable for the release
- **D.** Timely, specific recognition of the team in a way the team values

243. In a high-performing team, a member proposes a radically different design in a planning session. What should the leader encourage?

- **A.** Avoiding the topic so the team does not enter a conflict
- **B.** Constructive debate focused on the merits of the idea
- **C.** Rejecting the idea quickly to protect the planned schedule
- **D.** Voting on the idea immediately without further discussion

244. A project manager learns that a former colleague is applying for a role on her team. She has a close personal relationship with him. What should she do?

- **A.** Reject his application without review to avoid any appearance of bias
- **B.** Keep the relationship private and evaluate him using the normal criteria
- **C.** Disclose the relationship and let others evaluate the candidate
- **D.** Hire him directly, since she already knows he is a reliable worker

Answers — Part 11 (Leading People and Ethics)

Q220 — Answer: C

Correct: C. Directing — high direction and low support

An enthusiastic beginner (high commitment, low competence) benefits from clear direction.

Q221 — Answer: B

Correct: B. Transformational

Transformational leaders inspire change through vision and empowerment; transactional leaders manage by rewards and exceptions.

Q222 — Answer: A

Correct: A. Expert power

Expert power comes from knowledge and skills. Referent power comes from personal respect and admiration.

Q223 — Answer: C

Correct: C. Theory Y

Theory Y assumes people are self-motivated; Theory X assumes they must be controlled.

Q224 — Answer: D

Correct: D. Safety

Job security is a safety need; higher-level motivators such as recognition are less effective until it is addressed.

Q225 — Answer: B

Correct: B. Involve their functional managers or HR according to organisational policy

Escalation through appropriate channels is the next step after direct facilitation fails.

Q226 — Answer: D

Correct: D. Compromise/reconcile

Compromise involves mutual concessions; collaboration seeks a win-win solution that fully meets everyone's needs.

Q227 — Answer: D

Correct: D. Schedules

Schedules, priorities and resources are the most frequent sources; personality conflicts are among the least frequent root causes.

Q228 — Answer: A

Correct: A. Self-regulation

Self-regulation is managing one's own emotions and impulses.

Q229 — Answer: A

Correct: A. Discuss their concerns before finalising the decision

Low counts signal reservations that should be explored to build real consensus.

Q230 — Answer: C

Correct: C. BATNA

The best alternative to a negotiated agreement defines the walk-away point in the negotiation.

Q231 — Answer: D

Correct: D. Decline the offer, follow company policy and disclose it as appropriate

Accepting gifts that could influence a decision creates a conflict of interest. The Code of Ethics requires fairness and transparency.

Q232 — Answer: A

Correct: A. Address it with the team member, correct the records and follow the organisation's procedures

Honesty and responsibility require addressing the misconduct, correcting the information and following organisational procedures.

Q233 — Answer: C

Correct: C. It sets aspirational and mandatory standards for responsibility, respect, fairness and honesty

The Code defines four values with aspirational and mandatory standards for members and certification holders.

Q234 — Answer: D

Correct: D. Show empathy and explore flexible arrangements while agreeing realistic commitments

Emotional intelligence and servant leadership mean understanding causes and finding workable solutions, balancing the person's needs with the project's.

Q235 — Answer: B

Correct: B. Performing

Performing teams are highly functional and self-managing; norming teams are still establishing habits and trust.

Q236 — Answer: D

Correct: D. Mastery

Pink's three intrinsic motivators are autonomy (self-direction), mastery (getting better) and purpose (meaning).

Q237 — Answer: B, E

Correct: B. Treating mistakes as learning opportunities in retrospectives / E. Thanking people who raise problems early

Psychological safety grows when raising issues is welcomed and mistakes are treated as learning, not punished.

Q238 — Answer: D

Correct: D. Facilitate meetings so everyone contributes, for example by round-robin or written input

Inclusive facilitation ensures diverse perspectives are heard, improving decisions.

Q239 — Answer: A

Correct: A. Management focuses on processes and control; leadership on vision and people

Project managers need both: managing work and leading people.

Q240 — Answer: A

Correct: A. Refuse, follow the law and company policy, and seek legitimate ways to speed up the permit

Legal and policy requirements take precedence over local customs; using intermediaries is also prohibited.

Q241 — Answer: D

Correct: D. OSCAR

OSCAR is a coaching model; ADKAR is a change model; DMAIC is a Six Sigma method; SWOT is an analysis technique.

Q242 — Answer: D

Correct: D. Timely, specific recognition of the team in a way the team values

Effective recognition is timely, specific, sincere and suited to the people and culture.

Q243 — Answer: B

Correct: B. Constructive debate focused on the merits of the idea

Healthy conflict about ideas improves solutions, provided it remains respectful and focused on issues.

Q244 — Answer: C

Correct: C. Disclose the relationship and let others evaluate the candidate

The Code of Ethics requires disclosure of conflicts of interest and fairness in hiring decisions.

Part 12 — Business Environment, Tailoring and Toolbox

245. Which activity should come first when planning project compliance?

- **A.** Identify the applicable legal, regulatory, contractual and policy requirements with experts
- **B.** Schedule the final compliance audit, so that the team knows the date it is working toward from the start
- **C.** Assign compliance to a junior analyst so that senior staff can focus on delivery
- **D.** Request exemptions from the regulations that are likely to slow the project down

246. An internal audit finds that the project has not been storing personal data according to the company's data protection policy. What should the project manager do?

- **A.** Record the finding in the lessons learned register and continue with the current practice
- **B.** Ask the auditor to treat the finding as an observation rather than as a non-compliance
- **C.** Stop the non-compliant practice, inform the right stakeholders, correct it and prevent recurrence
- **D.** Plan to fix the issue during the final phase, when the team will have more of its capacity available

247. Which change model focuses on the psychological transition people experience: endings, the neutral zone and new beginnings?

- **A.** Bridges' transition model
- **B.** Satir change model
- **C.** ADKAR
- **D.** Kotter's 8 steps

248. In the Satir change model, what typically happens right after a "foreign element" is introduced?

- **A.** Chaos, with a temporary drop in performance
- **B.** A return to the late status quo
- **C.** A new status quo at a higher level of performance
- **D.** Immediate improvement in team performance

249. A new ERP system will change how 3,000 employees work. Which practice best supports adoption?

- **A.** Announce the change in a single email to all staff on the go-live date
- **B.** Leave adoption activities to the IT department, which owns the new system
- **C.** Identify change champions, provide training and communicate the reasons repeatedly
- **D.** Switch off the old systems without notice, so that all users have no choice but to adopt the new one

250. Which of the following best describes a value delivery office (VDO)?

- **A.** A finance unit that approves project budgets and tracks the value of each investment
- **B.** A procurement unit that negotiates agile contracts with external service providers
- **C.** A PMO that coaches teams and supports adaptive ways of working with a focus on value
- **D.** A PMO that concentrates on auditing projects for compliance with standard templates

251. Which external analysis technique examines political, economic, social, technological, legal and environmental factors?

- **A.** MoSCoW
- **B.** RACI
- **C.** SWOT
- **D.** PESTLE

252. A competitor launches a similar product earlier than expected. What should the project manager do first?

- **A.** Cancel the project immediately, since the competitor has already captured most of the market
- **B.** Analyse the impact on the business case and strategy with the sponsor and product owner
- **C.** Ask the team to work double shifts so the product can be launched as soon as possible
- **D.** Continue as planned, since the project scope and baselines have already been approved

253. Which practice best represents continuous improvement in an adaptive team?

- **A.** An annual audit of the team's process by the quality department
- **B.** A lessons learned workshop at the end of the project with all members
- **C.** Retrospectives each iteration that produce specific, tracked actions
- **D.** A new process tried every week so the team can compare as many as possible

254. Which statement about sustainability in projects is correct?

- **A.** It is the responsibility of the organisation's marketing and communications team
- **B.** It applies to construction and energy projects, but only rarely to software or service projects
- **C.** It mainly concerns recycling and reducing paper use in the project's offices
- **D.** It covers environmental, social and economic considerations across the product life cycle

255. An AI tool used to screen candidates for project roles seems to favour applicants from one university. What should the project manager do?

- **A.** Recruit from that university only, since the tool has identified the best candidates
- **B.** Stop relying on the tool for decisions, report the possible bias and ensure a fair human review
- **C.** Continue using the tool, since it is efficient and its results are completely objective by design
- **D.** Keep using the tool but do not mention the pattern, to avoid embarrassing the vendor

256. Which statement about AI-generated status reports is most appropriate?

- **A.** They replace the need for conversations with stakeholders about project status
- **B.** They should never be used, since stakeholders expect all reports to be written by people
- **C.** They can save time, but the project manager must verify them before distribution
- **D.** They need no review, since AI produces objective and consistent information

257. Which two items should an organisation establish when adopting AI tools for project work? (Choose two.)

- **A.** Policies defining approved tools and acceptable data use
- **B.** Elimination of human decision-making
- **C.** A rule that AI outputs never need checking
- **D.** Unrestricted use of confidential data in public tools

- **E.** Training to build AI literacy for project staff

258. What is the main purpose of tailoring the project approach?

- **A.** To give the project manager freedom from any organisational policies that slow down the project's work
- **B.** To make every project in the organisation identical so that they are easier to compare
- **C.** To add documentation to every project so that audits can be passed without difficulty
- **D.** To fit methods, processes and governance to the context so that value is delivered effectively

259. Which factor would most likely lead a team to tailor toward more frequent, smaller deliveries?

- **A.** Highly uncertain requirements where early user feedback is valuable
- **B.** A building foundation that cannot be modified after it has been poured
- **C.** A fully specified purchase of standard items from a catalogue
- **D.** A regulatory submission that must be complete before it can be reviewed

260. A project team must choose an artifact to show the probability and impact ratings of their top risks visually. Which artifact is most suitable?

- **A.** Heat map / probability and impact matrix
- **B.** Milestone list
- **C.** WBS dictionary
- **D.** Team charter

261. Which artifact would best help an executive understand where a product is heading over the next four quarters?

- **A.** Issue log
- **B.** Activity list
- **C.** Sprint burndown
- **D.** Product roadmap

262. Which statement about benefit owners is correct?

- **A.** They approve each work package before the team starts working on it in the project
- **B.** They monitor and realise benefits, often after the project has closed
- **C.** They are responsible for maintaining the project schedule baseline
- **D.** They are always members of the project team who report to the PM

263. Users of a new system have learned how to use it but continue reverting to old habits after a month. According to ADKAR, which element needs attention?

- **A.** Desire
- **B.** Awareness
- **C.** Knowledge
- **D.** Reinforcement

264. Which type of PMO directly manages projects and to which project managers report?

- **A.** Supportive
- **B.** Virtual
- **C.** Controlling

- **D.** Directive

265. During execution, the team discovers that a key impediment requires approval from another department's director. The team cannot resolve it. What should the project manager do?

- **A.** Engage the director, explain the impact and options, and escalate through governance if needed
- **B.** Bypass the approval step so that the team can keep working while the request is pending
- **C.** Remove the dependent work from scope without consulting the product owner or sponsor
- **D.** Tell the team to wait until the director notices the problem and resolves it on their own initiative

266. Which indicator is a leading indicator of the future benefit "increased online sales"?

- **A.** Number of users completing the new checkout in the pilot
- **B.** Total online sales revenue reported for the whole of the next year
- **C.** Number of contracts closed at the end of the project
- **D.** Final project cost compared with the approved budget

267. The PMO requires a 30-page status report every week for a small, two-month internal project. What is the most appropriate action?

- **A.** Propose a lighter, tailored reporting format to the PMO that still meets governance needs
- **B.** Produce the full report by copying content from previous weeks to save effort
- **C.** Stop producing the report without telling the PMO, since the project is small
- **D.** Escalate the PMO's requirement to the chief executive as an example of excessive bureaucracy

268. What is the main benefit of documenting tailoring decisions?

- **A.** Transparency about why the approach was chosen, so it can be reviewed and improved
- **B.** Evidence for external auditors only, since internal stakeholders rarely read it
- **C.** Protection against future changes, since documented decisions cannot be revisited by anyone
- **D.** A longer project plan, which is generally viewed favourably by governance bodies

269. Which external change would most directly require the project to revisit its compliance plan?

- **A.** A team member's holiday
- **B.** A change in the colour of the company logo on the product's packaging
- **C.** A new safety regulation affecting the product's design
- **D.** A new meeting room booking system

Answers — Part 12 (Business Environment, Tailoring and Toolbox)

Q245 — Answer: A

Correct: A. Identify the applicable legal, regulatory, contractual and policy requirements with experts

Compliance planning begins with identifying requirements; then they are classified, planned, assigned, monitored and audited.

Q246 — Answer: C

Correct: C. Stop the non-compliant practice, inform the right stakeholders, correct it and prevent recurrence

Non-compliance must be addressed immediately and transparently, with corrective and preventive actions.

Q247 — Answer: A

Correct: A. Bridges' transition model

Bridges distinguishes the external change from the internal transition people go through.

Q248 — Answer: A

Correct: A. Chaos, with a temporary drop in performance

The foreign element causes chaos; after a transforming idea, practice and integration lead to a new status quo.

Q249 — Answer: C

Correct: C. Identify change champions, provide training and communicate the reasons repeatedly

Organisational change succeeds with sponsorship, champions, communication, training and reinforcement.

Q250 — Answer: C

Correct: C. A PMO that coaches teams and supports adaptive ways of working with a focus on value

A VDO is a PMO adapted to agile environments, focused on enabling value delivery rather than enforcing process.

Q251 — Answer: D

Correct: D. PESTLE

PESTLE is a prompt list for scanning the external environment.

Q252 — Answer: B

Correct: B. Analyse the impact on the business case and strategy with the sponsor and product owner

External changes may affect value; the PM assesses impact and supports decision-makers with options.

Q253 — Answer: C

Correct: C. Retrospectives each iteration that produce specific, tracked actions

Frequent, actionable retrospectives embed continuous improvement in the team's rhythm.

Q254 — Answer: D

Correct: D. It covers environmental, social and economic considerations across the product life cycle

Sustainability spans the triple bottom line and all project areas, from sourcing to design to team well-being.

Q255 — Answer: B

Correct: B. Stop relying on the tool for decisions, report the possible bias and ensure a fair human review

AI can reproduce bias; responsible use requires human oversight, fairness and transparency.

Q256 — Answer: C

Correct: C. They can save time, but the project manager must verify them before distribution

AI augments the PM's work; accountability for accuracy and appropriateness stays with people.

Q257 — Answer: A, E

Correct: A. Policies defining approved tools and acceptable data use / E. Training to build AI literacy for project staff

AI governance and training enable responsible, effective use; unchecked outputs and uncontrolled data use create risk.

Q258 — Answer: D

Correct: D. To fit methods, processes and governance to the context so that value is delivered effectively

Tailoring adapts the approach to the context while respecting mandatory requirements.

Q259 — Answer: A

Correct: A. Highly uncertain requirements where early user feedback is valuable

Uncertainty and the value of feedback favour incremental, adaptive delivery.

Q260 — Answer: A

Correct: A. Heat map / probability and impact matrix

A heat map or probability-impact matrix displays risk ratings visually for prioritisation.

Q261 — Answer: D

Correct: D. Product roadmap

Roadmaps communicate the high-level direction and planned releases over time.

Q262 — Answer: B

Correct: B. They monitor and realise benefits, often after the project has closed

Benefit owners, often in the business, track and realise benefits during and after transition.

Q263 — Answer: D

Correct: D. Reinforcement

Reinforcement sustains change through recognition, measurement and corrective actions after implementation.

Q264 — Answer: D

Correct: D. Directive

Directive PMOs exercise high control by managing projects directly.

Q265 — Answer: A

Correct: A. Engage the director, explain the impact and options, and escalate through governance if needed

Servant leaders remove impediments beyond the team's control, using governance channels when necessary.

Q266 — Answer: A

Correct: A. Number of users completing the new checkout in the pilot

Leading indicators predict future outcomes; pilot adoption suggests future sales, which will be measured later as lagging indicators.

Q267 — Answer: A

Correct: A. Propose a lighter, tailored reporting format to the PMO that still meets governance needs

Tailoring for the organisation involves agreeing proportionate governance with the PMO.

Q268 — Answer: A

Correct: A. Transparency about why the approach was chosen, so it can be reviewed and improved

Documented tailoring supports alignment, governance and continuous improvement.

Q269 — Answer: C

Correct: C. A new safety regulation affecting the product's design

Regulatory changes directly affect compliance requirements and must be assessed and incorporated.

Part 13 — Calculations Workout

270. BAC = $120,000. The project is planned to take 12 months at an even rate. After 6 months, 40% of the work is complete and AC = $54,000. What is the CV? (In dollars, include the sign.)

(Fill in the blank.)

Answer: _____

271. Using the same data (BAC $120,000, 6 of 12 months elapsed, 40% complete, AC $54,000), what is the SPI?

- **A.** 0.40
- **B.** 0.80
- **C.** 0.89
- **D.** 1.25

272. Using the same data, what is the EAC if current cost performance continues?

- **A.** $120,000
- **B.** $126,000
- **C.** $135,000
- **D.** $150,000

273. Using the same data (BAC $120,000, EV $48,000, AC $54,000), what is the TCPI to complete within the BAC?

- **A.** 0.92
- **B.** 1.00
- **C.** 1.09
- **D.** 1.25

274. Three-point estimates for a task: O = 3 days, M = 6 days, P = 15 days. What are the PERT estimate and standard deviation?

- **A.** 8 days and 2 days
- **B.** 6 days and 3 days
- **C.** 7 days and 2 days
- **D.** 7 days and 1.5 days

275. Two independent sequential activities have PERT standard deviations of 3 days and 4 days. What is the standard deviation of the path?

- **A.** 3.5 days
- **B.** 5 days
- **C.** 7 days
- **D.** 12 days

276. A team grows from 5 to 9 people. By how many do the communication channels increase?

- **A.** 4
- **B.** 10

- **C.** 26
- **D.** 36

277. A risk has a 15% probability and an impact of −120, 000.*Anopportunityhasa*2540,000. What is the net EMV?

- **A.** +$10,000
- **B.** −$8,000
- **C.** −$28,000
- **D.** −$18,000

278. Option 1: develop in-house for $150,000 with a 20% chance of $50,000 extra cost. Option 2: outsource for $165,000 with a 10% chance of $10,000 extra cost. Which option has the lower expected cost and what is it?

- **A.** Option 1 at $160,000
- **B.** Option 2 at $165,000
- **C.** Option 1 at $150,000
- **D.** Option 2 at $166,000

279. What is the present value of $133,100 received in 3 years at a 10% discount rate?

- **A.** $93,170
- **B.** $100,000
- **C.** $110,000
- **D.** $121,000

280. An investment of $90,000 yields $20,000 in year 1, $30,000 in year 2 and $40,000 in year 3. What is the simple payback period?

- **A.** 2 years
- **B.** 2.5 years
- **C.** 3 years
- **D.** 4.5 years

281. FPIF contract: target cost $300,000, target profit $30,000, target price $330,000, ceiling price $360,000, buyer/seller share 80/20, actual cost $320,000. What is the final price?

- **A.** $330,000
- **B.** $346,000
- **C.** $350,000
- **D.** $360,000

282. Using the same FPIF contract (target cost $300,000, target price $330,000, ceiling $360,000, share 80/20), what is the point of total assumption?

- **A.** $330,000
- **B.** $337,500
- **C.** $345,000
- **D.** $360,000

283. A Kanban system has an average WIP of 18 items and a throughput of 6 items per week. What is the average cycle time in weeks?

(Fill in the blank.)

Answer: _____

284. Network: Start → A (5) → C (4) → End; Start → B (3) → D (8) → End; C also depends on B. What is the project duration?

- **A.** 9 days
- **B.** 11 days
- **C.** 12 days
- **D.** 20 days

Answers — Part 13 (Calculations Workout)

Q270 — Answer: -$6,000

EV = 40% × 120,000 = 48,000. CV = EV − AC = 48,000 − 54,000 = −$6,000 (over budget).

Q271 — Answer: B

Correct: B. 0.80

PV = 50% × 120,000 = 60,000. SPI = EV / PV = 48,000 / 60,000 = 0.80.

Q272 — Answer: C

Correct: C. $135,000

CPI = 48,000 / 54,000 = 0.889. EAC = 120,000 / 0.889 = $135,000.

Q273 — Answer: C

Correct: C. 1.09

TCPI = (BAC − EV) / (BAC − AC) = 72,000 / 66,000 ≈ 1.09.

Q274 — Answer: C

Correct: C. 7 days and 2 days

PERT = (3 + 24 + 15) / 6 = 7 days. σ = (15 − 3) / 6 = 2 days.

Q275 — Answer: B

Correct: B. 5 days

Add variances, not standard deviations: 9 + 16 = 25; √25 = 5 days.

Q276 — Answer: C

Correct: C. 26

5 people: 10 channels; 9 people: 36 channels; increase = 26.

Q277 — Answer: B

Correct: B. −$8,000

Threat EMV = −18,000; opportunity EMV = +10,000; net = −$8,000.

Q278 — Answer: A

Correct: A. Option 1 at $160,000

Option 1: 150,000 + 0.2 × 50,000 = 160,000. Option 2: 165,000 + 0.1 × 10,000 = 166,000. Option 1 is cheaper.

Q279 — Answer: B

Correct: B. $100,000

PV = 133,100 / 1.1^3 = 133,100 / 1.331 = $100,000.

Q280 — Answer: C

Correct: C. 3 years

Cumulative inflows: 20,000 after year 1, 50,000 after year 2, 90,000 after year 3. Payback occurs at 3 years.

Q281 — Answer: B

Correct: B. $346,000

Overrun = $20,000; seller absorbs 20% = $4,000; profit = 30,000 − 4,000 = $26,000; price = 320,000 + 26,000 = $346,000, which is below the ceiling.

Q282 — Answer: B

Correct: B. $337,500

PTA = (360,000 − 330,000) / 0.8 + 300,000 = 37,500 + 300,000 = $337,500.

Q283 — Answer: 3

Little's Law: cycle time = WIP / throughput = 18 / 6 = 3 weeks.

Q284 — Answer: B

Correct: B. 11 days

Paths: A-C = 5 + 4 = 9; B-D = 3 + 8 = 11; B-C = 3 + 4 = 7 (C starts after the later of A and B, at day 5, finishing at 9). The longest path is B-D at 11 days.

Part 14 — Integrated Scenarios

285. You take over a hybrid project midway. The previous project manager left no handover notes. The team uses sprints for software while a vendor builds hardware under a fixed-price contract. What should you do first?

- **A.** Re-plan the whole project from scratch so that the new plan reflects your own approach and assumptions
- **B.** Ask the team to move to a fully predictive approach so that the whole project is managed in one way
- **C.** Renegotiate the vendor contract as time and materials to gain flexibility for the remaining hardware work
- **D.** Review the plan, contracts, backlog, risk register and issue log, and meet the team and key stakeholders

286. A product owner wants to add a regulatory reporting feature mid-release. The release date is fixed by law, and the team's capacity is full. What is the best approach?

- **A.** Re-prioritise the backlog with the product owner so the mandatory feature fits, deferring lower-value items
- **B.** Ask the authorities to move the legal deadline so that the team can deliver the feature in the next release
- **C.** Add the feature to the release and ask the team to work overtime until the release date is reached
- **D.** Reject the feature, since the release scope was agreed and cannot change once the release has started

287. During execution of a predictive construction project, the client asks for higher-grade windows. The change would add $80,000 and two weeks. The contingency reserve is $50,000 and is reserved for identified risks. What should the project manager do?

- **A.** Document the change with its impact and funding needs and submit it through change control
- **B.** Use the contingency reserve to pay for the windows, since the reserve exists for unplanned costs
- **C.** Approve the change directly, since the client is the one requesting and funding the upgrade
- **D.** Refuse the request, since the scope baseline cannot be modified once construction has started

288. Halfway through an agile project, velocity drops by 30% over two sprints. The team says they are spending a lot of time on unplanned production support. What should the project manager or scrum master do?

- **A.** Make the support work visible and agree with the team and product owner how to handle it
- **B.** Report the team's lower productivity to management so that it can be addressed formally
- **C.** Tell the team to stop doing production support so that development velocity can recover
- **D.** Lengthen the sprints from two to three weeks so that more points are completed per sprint

289. A key stakeholder requests weekly detailed reports, but the communications plan specifies monthly dashboards. Producing weekly reports would take one analyst two days per week. What should the project manager do?

- **A.** Decline the request, since the communications plan has been approved and is the baseline

- **B.** Ask the sponsor to tell the stakeholder that the monthly dashboard is all that is available
- **C.** Explore the stakeholder's underlying information needs and agree a proportionate solution
- **D.** Produce the weekly reports, since stakeholder satisfaction justifies the analyst's effort

290. A vendor informs the project that a critical component will be four weeks late because of a supplier fire. The contract includes a force majeure clause. What should the project manager do first?

- **A.** Terminate the contract for cause, since the vendor has failed to meet the agreed date
- **B.** Apply the liquidated damages clause, since the delay is clearly the vendor's responsibility
- **C.** Wait for the component quietly, since informing stakeholders would only create concern
- **D.** Assess the schedule impact, review the contract terms and explore options with the vendor

291. The team is about to deploy a new system when a tester finds a security vulnerability. The sponsor wants to go live on schedule. What should the project manager do?

- **A.** Go live as planned and fix the vulnerability in the next release, without informing anyone about it
- **B.** Postpone go-live indefinitely until every possible vulnerability has been eliminated
- **C.** Ask the tester to reclassify the finding as minor so that the go-live can proceed
- **D.** Inform the sponsor of the vulnerability and its impact and recommend options per security policy

292. A team member uses a public generative AI service to write code and pastes in proprietary source code from the client. What should the project manager do?

- **A.** Ban all software development tools on the project to avoid any further incidents
- **B.** Take no action, since the generated code worked and the client has not complained
- **C.** Encourage the practice, since using AI to write code improves the team's productivity and morale
- **D.** Stop the practice, assess the exposure with security and legal teams and reinforce AI policies

293. At a phase gate, the business case shows that expected benefits have fallen below the investment cost because of a market change. What is the most appropriate recommendation?

- **A.** Continue the project, since a large part of the budget has already been spent
- **B.** Present the updated business case and options, including termination, to the steering committee
- **C.** Keep the updated figures confidential, so that the team remains motivated until the end of the project
- **D.** Expand the scope to increase the benefits, without changing the approved business case

294. In a matrix organisation, a functional manager reassigns a key developer to another project without warning. What should the project manager do first?

- **A.** Discuss the impact with the functional manager and look for a solution together
- **B.** Stop the affected work and wait until a new developer is assigned by the organisation
- **C.** Tell the developer to keep working on your project regardless of the reassignment
- **D.** Escalate the matter straight to the CEO so that the decision is reversed quickly

295. A distributed team spanning four time zones complains that decisions are made in meetings some members cannot attend. What should the project manager do?

- **A.** Stop holding meetings and rely on written status updates sent by email once a week
- **B.** Ask members in inconvenient time zones to leave the team and replace them with local people
- **C.** Agree inclusive working agreements with the team, such as rotating meeting times and recorded decisions

- **D.** Keep the meetings at the times that suit the project manager, since the PM coordinates the whole team

296. The organisation adopts a new sustainability policy requiring carbon reporting for all projects. Your project is in execution. What should you do?

- **A.** Let the sustainability team estimate the project's emissions alone, without involving the project team
- **B.** Assess the new requirements, plan data collection and reporting, and process changes through governance
- **C.** Pause the project until the end of the year, when the policy's reporting cycle will begin
- **D.** Ignore the policy, since the project started before it was introduced and has an approved plan

297. Two senior stakeholders disagree on the project's primary objective: one wants speed to market and the other wants maximum features. What should the project manager do?

- **A.** Choose speed to market, since it is usually the better choice in competitive markets
- **B.** Try to satisfy both by adding all of the features and also accelerating the delivery schedule
- **C.** Facilitate a session using the business case and strategy, escalating to the sponsor if needed
- **D.** Wait for the stakeholders to resolve the disagreement between themselves

298. A risk owner reports that the trigger for an identified supplier risk has occurred, but the planned response is no longer feasible because the alternative supplier closed. What should happen?

- **A.** Use the fallback plan if one exists, or a workaround, then update the risk register
- **B.** Take no action, since the risk was identified in advance and the response has already been planned
- **C.** Close the risk in the register, since its trigger has now occurred and passed
- **D.** Transfer the risk to the team, which is now responsible for finding a supplier

299. A new team member joins a high-performing agile team. Shortly afterwards, the team shows signs of conflict and slower decisions. What is the most likely explanation and action?

- **A.** The team should stop holding retrospectives until the new member is fully integrated into the team
- **B.** The team may have regressed to an earlier stage; support onboarding and revisit team agreements
- **C.** The conflict is unrelated to the new member, so no particular action is needed
- **D.** The new member is not competent enough and should be replaced as soon as possible

300. The CPI of a project is 0.75 and the SPI is 1.05. The sponsor asks for the main concern. What is the best answer?

- **A.** Schedule: the project is behind and needs to be fast-tracked urgently
- **B.** Costs: the project is well over budget while slightly ahead of schedule
- **C.** Neither: both indices are within the normal range for a project at this stage
- **D.** Quality: a CPI below 1 shows that the defect rate is above the agreed target

301. An agile team discovers that the definition of done does not include accessibility testing, although the product must meet legal accessibility standards. What should be done?

- **A.** Wait until users complain about accessibility, and then decide whether any changes are worth making
- **B.** Remove accessibility from the requirements, since it was not in the definition of done
- **C.** Add accessibility checks to the definition of done and put the gaps in completed work on the backlog
- **D.** Ask users to report accessibility problems after release so they can be fixed in support

302. During closure, the operations team refuses to accept the system because support documentation is missing. What should the project manager do?

- **A.** Hand the documentation task to the operations team, since they will use the system
- **B.** Complete the missing transition deliverables required by the acceptance criteria before closing
- **C.** Close the project anyway, since the system itself has been delivered and tested successfully by the team
- **D.** Ask the sponsor to instruct operations to accept the system without the documentation

303. A project in a government agency must comply with strict procurement rules, but the team wants to bring in a specialist consultant within days. What is the most appropriate action?

- **A.** Engage the consultant informally and reimburse the fees through the team's expense claims
- **B.** Ask the consultant to start working for free until the formal contract has been signed
- **C.** Work with procurement to use an approved, compliant route such as an existing framework agreement
- **D.** Skip the procurement rules this once, since the business need is urgent and the request is well justified

304. A sponsor asks the project manager to commit to a delivery date before any estimates exist. What is the best response?

- **A.** Refuse to discuss any dates until the requirements and design have been fully completed
- **B.** Explain the uncertainty, give a range and agree when a more reliable estimate will be available
- **C.** Propose an aggressive date to motivate the team and adjust it later if needed
- **D.** Give the date that the sponsor would like, since meeting the sponsor's expectations is the top priority

Answers — Part 14 (Integrated Scenarios)

Q285 — Answer: D

Correct: D. Review the plan, contracts, backlog, risk register and issue log, and meet the team and key stakeholders

A new PM first builds understanding from existing artifacts and people before changing anything.

Q286 — Answer: A

Correct: A. Re-prioritise the backlog with the product owner so the mandatory feature fits, deferring lower-value items

In adaptive delivery with fixed time, scope is traded based on value and obligations; mandatory compliance items take priority.

Q287 — Answer: A

Correct: A. Document the change with its impact and funding needs and submit it through change control

A scope change is not a risk event, so contingency reserve should not fund it. The change goes through integrated change control with its impact on cost and schedule.

Q288 — Answer: A

Correct: A. Make the support work visible and agree with the team and product owner how to handle it

Transparency and joint problem-solving address the root cause of the drop while respecting real operational needs.

Q289 — Answer: C

Correct: C. Explore the stakeholder's underlying information needs and agree a proportionate solution

Understanding needs and negotiating a proportionate solution balances stakeholder satisfaction with project effort.

Q290 — Answer: D

Correct: D. Assess the schedule impact, review the contract terms and explore options with the vendor

Impact assessment and contract review come first; force majeure may excuse the delay, so penalties may not apply.

Q291 — Answer: D

Correct: D. Inform the sponsor of the vulnerability and its impact and recommend options per security policy

Transparency, compliance with security policy and accountable leadership require informing the decision-maker with options.

Q292 — Answer: D

Correct: D. Stop the practice, assess the exposure with security and legal teams and reinforce AI policies

Confidentiality and responsible AI use require immediate containment, assessment and reinforcement of policy.

Q293 — Answer: B

Correct: B. Present the updated business case and options, including termination, to the steering committee

Focusing on value means decision-makers should know when the business case no longer holds; sunk costs are irrelevant.

Q294 — Answer: A

Correct: A. Discuss the impact with the functional manager and look for a solution together

Direct negotiation with the resource manager is the first step; escalation follows if the conflict cannot be resolved.

Q295 — Answer: C

Correct: C. Agree inclusive working agreements with the team, such as rotating meeting times and recorded decisions

Inclusive working agreements ensure every member can participate in decisions.

Q296 — Answer: B

Correct: B. Assess the new requirements, plan data collection and reporting, and process changes through governance

New organisational requirements are assessed and incorporated through governance, consistent with the sustainability principle.

Q297 — Answer: C

Correct: C. Facilitate a session using the business case and strategy, escalating to the sponsor if needed

Aligning expectations requires facilitation grounded in the business case; the sponsor decides if they cannot agree.

Q298 — Answer: A

Correct: A. Use the fallback plan if one exists, or a workaround, then update the risk register

When the primary response fails, a fallback plan or a workaround is used, followed by updates to risk documentation.

Q299 — Answer: B

Correct: B. The team may have regressed to an earlier stage; support onboarding and revisit team agreements

Membership changes often cause teams to revisit earlier stages; onboarding and refreshing agreements help them return to performing.

Q300 — Answer: B

Correct: B. Costs: the project is well over budget while slightly ahead of schedule

CPI below 1 means overspending; SPI above 1 means slightly ahead. The cost overrun is the priority.

Q301 — Answer: C

Correct: C. Add accessibility checks to the definition of done and put the gaps in completed work on the backlog

Compliance requirements should be embedded in the DoD; existing gaps are added to the backlog and prioritised.

Q302 — Answer: B

Correct: B. Complete the missing transition deliverables required by the acceptance criteria before closing

Transition readiness is part of the deliverables; closure requires formal acceptance by the receiving organisation.

Q303 — Answer: C

Correct: C. Work with procurement to use an approved, compliant route such as an existing framework agreement

Compliance with procurement rules is mandatory; approved mechanisms may provide speed without breaking rules.

Q304 — Answer: B

Correct: B. Explain the uncertainty, give a range and agree when a more reliable estimate will be available

Honest communication of uncertainty with progressive elaboration supports good decisions and trust.